Exploring social behavior of the largest bioelectric capacitator via long-term electric recordings

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Contents

2 Abstract

Electric organ discharges (EODs) provide a non-invasive way to study electric eel behaviour over timescales that are difficult to capture through direct observation. This thesis investigates long-term patterns in EOD waveform and activity in captive *Electrophorus voltai* using continuous multi-electrode recordings from the Museum für Naturkunde Berlin. The dataset contains more than 13 million accepted EODs recorded between 2023 and 2026. Waveforms were classified with robust principal component analysis and a random forest into normal, wide, or double labels; half-width and PCA views show substantial overlap among these labels, consistent with a morphological continuum whose extremes remain clearly distinguishable. Pulse activity was analysed across multiple timescales and in relation to feeding and position within the tank. Overall pulse activity showed a clear feeding-related response, increasing from approximately 1.34 Hz outside feeding windows to 5.03 Hz during feeding. Wide-pulse activity showed a similar response, while double-pulse activity remained low and did not show a comparable increase in mean rate. Across the long-term record, the strongest increase in special-pulse activity occurred during a distinct period in late 2025 rather than as a recurring calendar-month pattern. This increase coincided with a shift in coarse space use toward the darker region of the tank and with unusual observations of increased proximity between the animals. Taken together, these changes are consistent with a change in social state and raise the possibility of a reproductive context, but the present data do not demonstrate a reproductive event or identify special pulses as reproductive signals. The results establish a long-term electrophysiological baseline for captive electric eels and provide an analytical framework for future comparison with recordings from the natural environment and future captive breeding endeavors.

3 Zusammenfassung

Elektrische Organentladungen (EODs) bieten eine nicht-invasive Möglichkeit, das Verhalten von Zitteraalen über Zeitskalen hinweg zu untersuchen, die durch direkte Beobachtung nur schwer zu erfassen sind. Diese Arbeit untersucht Langzeitmuster in den EOD-Wellenformen und der Aktivität von in Gefangenschaft gehaltenen *Electrophorus voltai* anhand kontinuierlicher Mehr-Elektroden-Aufzeichnungen aus dem Museum für Naturkunde Berlin. Der Datensatz umfasst mehr als 13 Millionen akzeptierte EODs, die zwischen 2023 und 2026 aufgezeichnet wurden. Die Wellenformen wurden mithilfe einer Kombination aus robuster Hauptkomponentenanalyse und Random-Forest-Klassifizierung klassifiziert, und die Pulsaktivität wurde über mehrere Zeitskalen hinweg sowie in Bezug auf die Fütterung und die Position im Becken analysiert. Es wurden drei reproduzierbare Wellenformklassen identifiziert: normale, breite und Doppelpulse. Die Gesamtimpulsaktivität zeigte eine deutliche fütterungsbezogene Reaktion und stieg von etwa 1,34 Hz außerhalb der Fütterungsfenster auf 5,03 Hz während der Fütterung an. Die Aktivität der breiten Pulse zeigte eine ähnliche Reaktion, während die Doppelpulsaktivität niedrig blieb und keinen vergleichbaren Anstieg der mittleren Frequenz aufwies. Über die gesamte Langzeitaufzeichnung hinweg trat der stärkste Anstieg der Spezialpulsaktivität während eines bestimmten Zeitraums Ende 2025 auf und nicht als wiederkehrendes Muster im Kalendermonat. Dieser Anstieg fiel mit einer Verschiebung der grobskaligen Raumnutzung in Richtung des dunkleren Bereichs des Beckens sowie mit ungewöhnlichen Beobachtungen einer erhöhten Nähe zwischen den Tieren zusammen. Insgesamt stehen diese Veränderungen im Einklang mit einer Veränderung des Sozialzustands und lassen die Möglichkeit eines Fortpflanzungskontextes vermuten. Die vorliegenden Daten können ein Fortpflanzungsereignis nicht zureichend belegen, jedoch deuten sie auf eine mögliche Verbindung die weiter untersucht werden sollte. Die Ergebnisse bilden eine ausführliche elektrophysiologische Grundlage für in Gefangenschaft lebende Zitteraale und bieten einen analytischen Rahmen für zukünftige Vergleiche mit Aufzeichnungen aus der natürlichen Umgebung oder Paarungsversuche in Gefangenschaft.

4 Introduction

Electric eel electrocytes have been inspiring human engineering for longer than most people realize. The best known example is probably Volta's battery, which he described as "Artificial Electric Organ", was inspired by the electric organ of electric eels (Volta 1800). Since then, electric eels and their ability to generate electricity have inspired the development of modern electrochemical capacitors and iontronic artificial skins that claim better temperature and touch sensing than competing technologies (Pei et al. 2025; Sun et al. 2016). The animal itself still outruns most of that hardware. De Santana et al. (2019) recorded discharges of 860 V in *Electrophorus voltai*, making electric eels the strongest living bioelectricity generators known.

That alone would be a good enough reason to research electric eels. One could argue that *Electrophorus* is one of the most interesting animals on the planet for scientists across fields to study because it has many more unusual and unique traits. They are one of the few strongly electric fish genera in a modality otherwise studied mostly in weakly electric gymnotiforms. Although their natural habitats are remote, electric eels are freshwater fish, allowing to be studied in freshwater settings (De Santana et al. 2019; Mendes-Junior et al. 2016). Electric eels are obligate air-breathers, using their highly vascularized oral mucosa for efficient gas exchange (Johansen et al. 1968). This enables them to inhabit hypoxic waters with oxygen levels below 1 mg/L (Bárbara et al. 2010). Their adaptation to this environment and ability to generate electricity has made them an interesting model organism for evolutionary biologists. Ecologically, electric eels are very interesting since they are apex predators in their food web and have no natural enemies other than humans. The most interesting trait for ethologists is probably the surprisingly complex social behavior of *Electrophorus*. They are animals with extensive parental care, which is seldom in fish to begin with, and the care work is taken on by both mating partners (Assunção and Schwassmann 1995; Bastos 2020). *E. voltai* have been reported to show complex hunting behavior suggesting a much more social behavior than the prevailing scienctific understanding (Bastos et al. 2021). Additionally, *Electrophorus* is one of the largest freshwater fish in the world (up to 2.5 m in length and 20 kg in mass), making them a great subject for the study of electric organs (Mendes-Junior et al. 2016). *Electrophorus* spp. possess three electric organs (main, Hunter's, and Sachs'), each with a different functional role (Xu et al. 2021). Laboratory work even suggests that eel shocks can facilitate DNA uptake in teleost larvae. This finding tentatively suggests that high voltage discharges contribute to genetic variability in the limnic south american ecosystems (Sakaki et al. 2023). For all that, surprisingly little is known about electric eels in the wild.

For most of taxonomic history there was one electric eel, *Electrophorus electricus*. According to newer phylogenetic studies by De Santana et al. (2019) there are three to four species, not just *electricus*. Preliminary double-digest restriction-site associated DNA (ddRAD) analyses of more than 300 individuals from across Greater Amazonia now recover two major ecotypes of *Electrophorus*: upland lineages of shield headwaters and rapids, and lowland lineages of whitewater and blackwater floodplains (de Santana 2025). Upland populations are deeply diverged across the Guiana and Brazilian shields, whereas lowland lineages show shallower genomic divergence that is nonetheless structured by water type. Despite that ecological and genetic partitioning, all lineages retain a conserved external morphology, so the currently named species still understate the hidden diversity. There is a distinct sexual dimorphism as males are significantly larger than females. The average male length was 124.2 cm and mean mass 3.62 kg, versus a mean length of 94.3 cm and 1.92 kg for females (Mendes-Junior et al. 2016). No range-wide population estimate is currently available for electric eels due to the sparse data. An adequate observation of conservation status is therefore not possible. Amazonian floodplain habitats are increasingly exposed to drought-associated warming and hypoxia Johannsson et al. (2026) and Röpke et al. (2022), deforestation Ilha et al. (2018), and water contaminations Viana et al. (2021). These stressors are likely to threaten electric-eel populations through habitat change. Even though the Amazonian habitats remain remote, human influence on electric eel populations is increasing (fishing, gold mining, and hunting by locals driven by fear).

Studying these animals in captivity is difficult since breeding depends on specialised nest sites and seasonal hydrological cues that are hard to recreate (Assunção and Schwassmann 1995; Bastos 2020). The sheer size of the animals requires subsantial efforts and funds to provide them with an adequat and even than is not comparable to the natural habitat which in turn likely influences the behavior of captive eels. Because electric eels are are crepuscular to nocturnal, and white and blackwater rivers such as the Rio Negro and Amazon are murky. So studying electric eels in the wild is equally complicated since visual observation is a poor primary tool. Field biology of electric eels therefore faces the dilemma of high conservation relevance and low accessibility.

Electric organ discharges (EODs) are the practical way in. Recording this modality comes with the advantage that it is non-invasive and since the EOD is the main sensory and communication channel for electric eels it is likely to yield the most information. Electric eels emit low-voltage EODs during search behaviour. Search pulse trains are irregular with an approximate pulse rate that changes with behavioral context to approximately 10–20 Hz during excitation and hunting (Catania 2019). Predatory strikes involve high-voltage volleys at much higher rates (Catania 2015b). According to Catania (2017), high-voltage scales roughly with body length, while girth mainly affects internal resistance and available current. Low voltage pulses are used for electrolocation of surroundings, conspecifics, and prey (Catania 2019; Keynes and Martins-Ferreira 1953). Generation of the discharge itself rests on the stacked electrocyte membrane potentials that Keynes and Martins-Ferreira (1953) measured directly from the organ of Sachs.

Hunting sequences make the voltage switch explicit. The eel first senses prey with low voltage pulses, then fires a high voltage double pulse that forces an involuntary twitch. That tactile confirmation, not electric feedback alone, is what releases the suction strike (Catania 2014). High voltage volleys then track and stun fast moving prey with striking spatial precision (Catania 2015a; Catania 2015b). *Electrophorus* is piscivorous. Prey mainly consists of fish, up to 20 cm, and can even contain Crustaceans (Bastos et al. 2021; Mendes-Junior et al. 2016). In addition to the two pulse forms established by Catania (2019), a third has been reported that may be rarer or context restricted (Xu et al. 2021). Passive electrode arrays can recover the behaviorally relevant voltage scale without the invasive recordings that older work required. Additionally, long-term recordings provide an opportunity to look beyond the established discharge categories and to identify waveform variation that may be rare, or difficult to detect during short recordings.

The highly unusual phenomenon of group hunting has recently first been documented in the field. Aggregations of more than 100 *E. voltai* herd small fishes into dense prey balls at dawn and dusk before subsets of two to ten individuals launch synchronized high-voltage strikes (Bastos et al. 2021). Those events imply social coordination that is still mostly known from anecdote. Outside hunting aggregations, whether electric eels are typically solitary, loosely associated, or seasonally gregarious remains open. Beyond the ethology, this finding has inspired Electric Eel Foraging Optimization algorithms used in energy-system engineering, including islanded microgrid control (Ebrahim et al. 2025; Zhao et al. 2024).

The knowledge and scientific basis for the reproduction of electric eels is thinner still. Mating triggers are poorly known. Armoured catfish offer a tentative comparative foothold for nest related cues (Hostache and Mol 1998), but the mapping is speculative. To our knowledge, electric eels have never been bred in captivity, so almost everything about courtship comes from field scraps (Assunção and Schwassmann 1995; Senarat et al. 2025). Adults migrate to find nest sites and raise young. Nests are built in small, clear tributaries near or in Igarapes (smaller, calmer, cooler, clearer sidearms) tucked into root masses at the stream margin and hard to see from outside. The male builds a bubble nest right below the surface using secretions from its mouth (Assunção and Schwassmann 1995; Bastos 2020). The female deposits eggs into it and largely remains inside the root mass of the nest around egg placement. The male then patrols and guards for an extended period (Assunção and Schwassmann 1995; Bastos 2020). Parental care lasts about four months (Bastos 2020), with shorter care reported for highland species (*E. voltai*, *E. electricus*) than for *E. varii*. Ovarian counts in *E. electricus* give a mean of 2 633 mature oocytes per female (Sá-Oliveira and Mendes Júnior 2012). Field estimates put the number of hatchlings shortly after spawning at about 300–400 per nest (Bastos 2020). Eggs from several females may end up in one nest, with males competing via nest quality. The multi female pattern is borrowed from catfish natural history (Hostache and Mol 1998) and remain an assumption for *Electrophorus*. However, it would explain the discrepant numbers of eggs per nest relative to the number of mature oocytes per female. Nesting with male parental care was long thought rare among gymnotiforms (Crampton and Hopkins 2005). Across fishes more broadly, nest forms are diverse and often linked to parental care, but also to sexual selection and refuge (Bessa et al. 2022). In electric eels it appears unusually long and behaviorally rich (Bastos 2020).

This lack of information on basic and crucial behaviours is the motivation for the approach used here. Passive electrode grids record the animals through their main sensory and communication channel, the EOD. The method does not require clear water or daylight, leaves the habitat largely undisturbed, and avoids handling stress that could alter the social behaviour under investigation. It is therefore well suited to long-term monitoring and to exploring whether changes in EOD waveform and activity are associated with different behavioural states.

The present thesis focuses on continuous multi-electrode recordings from captive *E. voltai* at the Museum für Naturkunde Berlin. The primary aim is to characterize the EODs recorded over several years, including unusual waveform classes, and to examine their occurrence across different timescales and behavioural contexts. In particular, pulse activity is examined in relation to feeding and coarse space use within the tank. At the same time we point out that the electrical patterns cannot by themselves be assigned to specific behavioural acts.

A second objective is to provide a dataset and analytical framework that can later be applied to recordings from the natural environment. Parallel field recordings of *E. voltai* were collected close to Macapá in the Floresta Nacional (FLONA), including the Rio Araguari and Rio Falsino, and in the Ducke reserve in Manaus. These recordings are not analysed quantitatively in the present thesis, but they provide an important basis for future comparison between captive and natural conditions. The captive recordings offer the duration and controlled context needed to identify long-term patterns. The purpose of the field recordings we obtained is to determine how the results presented here relate to the ecology and behaviour of *E. voltai* in its natural habitat.

By establishing a long-term electrophysiological baseline and identifying waveform and activity patterns that may be associated with behavioural states, this work contributes to the limited understanding of electric eel social behaviour and provides a basis for future studies under natural conditions.

5 Materials and Methods

This thesis combines hardware development, a four week field expedition to Brazil in September 2025, and long term captive recordings in Berlin. The quantitative Results use only the Berlin dataset. The field recordings are described here because they motivated much of the hardware work and provide the natural habitat counterpart for future analyses; they are not analysed quantitatively in this thesis.

Custom multi electrode loggers, foldable square grids, flexible electrode lines, waterproof housings, and flight compatible battery packs were built before departure (Section 5.1). The first two weeks of the expedition were spent in Manaus at Reserva Ducke and the Instituto Nacional de Pesquisas da Amazônia (INPA). At Ducke, grids and lines were deployed at four stream sites near known *Electrophorus* activity, including nest associated locations. At INPA, loggers were tested and calibrated in an enclosure with captive eels. The second two weeks were spent near Macapá in the Floresta Nacional (FLONA) do Amapá, with continuous deployments along the Rio Falsino and Rio Araguari. There, nine square grid (or tetrahedron) sites and seven line sites were instrumented in parallel, with batteries exchanged every few days (Figure 1).

(a) Surface floating 4×4 grid with bottle floats and logger housing at a Rio Falsino bank site.

(b) Submerged grid at a Rio Araguari site, viewed from a boat during deployment.

Figure 1: Electrode grids deployed in FLONA.

Table 1 summarises recording effort for the three datasets. Berlin provides the multi year captive baseline used throughout the Results. Ducke and FLONA together contribute about 11 TiB of field WAV data from up to 16 concurrent loggers; those recordings remain for follow up work that applies the same classification and activity pipeline under natural conditions.

Table 1: Recording inventory for the Berlin captive dataset and the September 2025 Brazil field campaigns. Logger hours are approximate, based on the number of stored five minute WAV segments.

Dataset	Location	Period	Loggers	Arrays / sites	Effort (raw size)
Berlin	MfN tank	Nov 2023–Mar 2026	1–2	16-ch line(s)	∼3770 h over 165 d (13.7 TiB)
Ducke	Reserva Ducke	1–2 Sep 2025	7	4 sites (grids + lines)	∼236 h (1.3 TiB)
FLONA	Falsino & Araguari	15–22 Sep 2025	16	9 grids/tetra + 7 lines	∼1970 h (9.9 TiB)

5.1 Recording hardware

Recordings used custom multi electrode loggers based on Teensy microcontrollers with 16 channel amplifier boards. Software and board designs are available at https://github.com/janscience/TeeGrid, https://github.com/janscience/TeeRec, and https://github.com/janscience/Teensy_Amp. Each device digitizes 16 electrode channels continuously to disk as multi channel WAV files.

Electrodes were wound by hand from stainless steel wire (three turns). Three array geometries were built: lines, square grids, and one tetrahedron for exploratory experiments.

Grids. Square frames with 180 cm side length carried 16 electrodes in a 4×4 layout at 60 cm spacing. Grid frames were assembled from PVC tubing: tubes form the edges, straight segments use sleeved plug-in PVC joints, and corners use PVC cross pieces with glued-on connectors. This construction was foldable for transport while remaining robust and quick to deploy; the packed frame folds to 60 cm edge length. For assembly, tubes were laid out in the target configuration, electrodes slipped onto the tubes, and cable runs cut with slack for joints plus about 150 cm from grid to housing. Stretchy ropes (four on inner bars, two on cross bars) tension the frame. Cables were soldered to the matching electrodes, sealed with waterproof heat shrink, and routed under spiral tubing along cable-bearing tubes.

Lines. Flexible electrode lines use a cable bundle wrapped in spiral tubing. Flexibility reduces mechanical damage in the field and simplifies packing relative to rigid frames. Total length is 15.6 m: the first electrode sits 150 cm from the housing end, and the remaining electrodes follow at 80 cm spacing, totalling 16 electrodes. For assembly, 16 cables were cut to length, bundled, and wrapped in spiral tubing with electrode positions marked. At each site the matching cable was pulled out of the wrap, the electrode wound onto tubing and cable, then soldered. Waterproof heat shrink sealed each joint. Spiral wrap was applied in inter-electrode segments with about 20 cm overlap at electrodes for stability. Tube diameter decreased toward the free end as fewer cables remained.

The cable to housing transition used a threaded gland. To ensure that the transition glans was water proof, heat shrink was applied over the 16 cable bundle. Then the gland volume was filled with epoxy resin, avoiding air bubbles and epoxy leakage that would stiffen the free cables or interfere with their fit in the housing.

5.1.1 Waterproofing and power

Solder joints were sealed with waterproof heat shrink. Electronics sat in PVC housings. These housings were waterproofed with threaded end caps and O-rings. Before closing, O-rings were cleaned and silicone paste was used to insure seamless sealing. The outside to inside cable interface was potted in epoxy as above.

Battery packs were built from seven Lithium ion cells wired in parallel via nickel sheet on like poles (positive to positive, negative to negative). A protection circuit was soldered on after corrosion protection spray. Contacts were kept from overheating during soldering by using minimal temperature and heat exposure time. Each pack was heat shrunk, capped with cut PVC plates, heat shrunk again, and labeled. Packs were stored at about 3.4 V rest charge. Capacity is 24.5 Ah per pack, which is within typical hand luggage flight limits, and lasts for about 60 h or 2.5 days of logger runtime.

5.2 Berlin captive recordings

The data that builds the basis of the analysis conducted in this thesis were recorded in collaboration with Peter Bartsch at the Museum für Naturkunde (Natural History Museum Berlin). Two adult eels, one male and one female, were housed in a dual-pool tank constructed from two pools of about 3 m diameter joined by an opening of about 2 m. One pool was kept as a brighter region and the other as a darker region of the tank. In the Berlin setting most recordings were obtained using a 16-electrode line with 25 cm spacing, totalling 3.75 m in length. Overall, the Berlin dataset covers session starts on 165 calendar days from November 2023 to March 2026 (Table 1), about 13.7 TiB of raw WAV and around 500 GiB of processed NIX/H5 data.

5.3 Video and audio synchronization

Where video was available, logger time and camera time were aligned with the custom open-source tool videosync. An LED on the logger blinks pseudo-randomly with a known schedule. videosync uses a video region of interest (ROI) to track the blinking LED on the logger. Blink timing is recovered from logger metadata. Then both traces are cross correlated to estimate the temporal offset. Highest cross correlation value is used to align the two time bases. Logger audio was band-pass filtered within 100 to 3000 Hz, then stitched across overlapping WAV segments and muxed onto the video for later behavioral scoring.

5.4 Data analysis

Analysis of the Berlin recordings has two main stages. First, EOD pulses are detected in the continuous WAV traces to obtain the time of every discharge and a short waveform cutout around it (Figure 2). Second, those pulses are analysed by waveform morphology and by discharge rate, as detailed in the subsections below.

5.4.1 Peak detection and EOD labeling

Continuous recordings were processed with eeltracker. The tool detects candidate pulses, groups them into multi-channel events with short waveform cutouts, and writes them to NIX/HDF5. Downstream analyses keep only events that have been classified as EODs. No amplitude threshold or high-voltage/volley filter is applied: low-voltage search pulses and high-voltage discharges (shocks) are retained alike whenever they pass that EOD acceptance step. In the Results and later analyses, phrases such as "all pulses" or "all EODs" refer to this full accepted set.

5.4.2 EOD waveform classification

Accepted EODs were further labelled by waveform morphology into three working classes: normal, wide, and double pulses (Figure 5). Classification uses peak-normalized waveforms, so labels reflect shape rather than absolute amplitude; a high-voltage pulse and a low-voltage pulse of the same shape receive the same class. Normal pulses are short single peaks while wide pulses are longer single peaks with a slower falling edge or shoulder. Double pulses show two peaks separated by a shallow trough (training labels also included clear shoulders toward a second peak). A seed set was labelled by hand

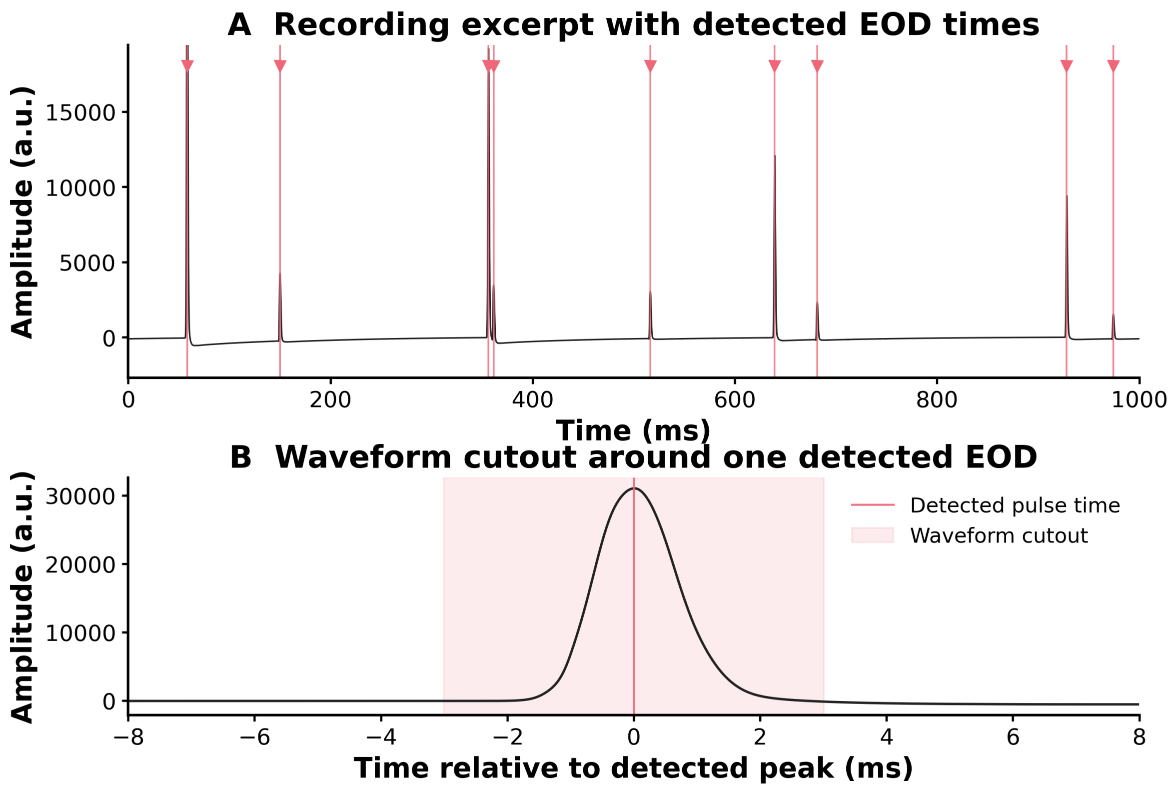


Figure 2: EOD pulse detection on a short Berlin recording excerpt (1 s, one electrode channel). Red markers in (A) show eeltracker pulse times. (B) zooms on one detected discharge and shades the short waveform cutout stored for later shape analysis.

according to these shapes. Waveforms were then reduced by RobustPCA (at most 20 components) and classified with a random forest trained on those labels. Rare classes (wide, double) were accepted only if their predicted probability was at least 0.40. The resulting markers (`is_double_peak`, `is_wide_pulse`, and `special_pulse_class`) provide the class assignments used throughout the Results. These discrete labels are an operational scheme for quantifying prevalence and rates rather than a claim of fully discrete natural morphologies.

Classified pulses were characterised by half-maximum width and by mean and median prototype waveforms per class; class prevalence is summarised over the dataset (Figures 4–6). To test whether the double shape is consistent with a short doublet of normal pulses, the median double prototype was fit as the sum of two median normal templates:

$$y(t) = a_1\, T_{w_1}(t - t_1) + a_2\, T_{w_2}(t - t_2), \tag{1}$$

where T_w is the median normal template time-scaled by width factor w ($w = 1$ keeps the native normal width; $w > 1$ widens), a_1 and a_2 are free amplitudes, and t_1 and t_2 are free peak times with $\Delta t = t_2 - t_1$. Six nested models (A–F) relaxed constraints on Δt and on the width factors (Table 2; Figure 9). Amplitudes a_1 and a_2 are free in every model. Model E was chosen as the working description: free a_1, a_2, t_1, and t_2, with a shared free width factor $w_1 = w_2 = w$. Allowing independent widths w_1 and w_2 in model F did not improve the fit meaningfully. Labeled pulses were also examined in Robust PCA space (Figure 7).

Table 2: Constraints and fitted amplitudes for the six nested two-normal fits of the median double pulse. Fixed Δt locks peak times to the observed double peaks; free Δt lets t_1 and t_2 vary. Width factor $w = 1$ is the native median-normal template; shared means $w_1 = w_2$. Amplitudes a_1 and a_2 are free in every model; the table lists the best-fit values on the median double.

Model	Δt	**Widths** w_1, w_2	**Fitted** a_1	**Fitted** a_2
A	free	fixed ($w_1 = w_2 = 1$)	0.69	0.78
B	fixed to peaks	fixed ($w_1 = w_2 = 1$)	0.63	0.78
C	fixed to peaks	shared, free	0.61	0.76
D	fixed to peaks	independent, free	0.62	0.75
E	free	shared, free	0.88	0.94
F	free	independent, free	0.92	0.90

5.4.3 Activity rates

Pulse times are binned at minute, hour, day, month, month since recording start, and year. Minute and day bins were discarded in the final analysis because they did not add information beyond the hour and month bins. Rates are counts divided by recording effort in each bin, separately for all EODs, double pulses, and wide pulses. The recording effort describes the total logger-on time (in seconds) that falls into that bin, not calendar time: a month with only a few recording days contributes only those recorded seconds. This converts raw pulse counts into rates in Hz and avoids treating more heavily sampled periods as more active. A second electrode line was added in November 2025 to span bright and dark regions of the tank. Dual-line recordings from 25 November 2025 onward enter rate analyses with pulse counts weighted by 0.5, so the second line does not double-count activity. The multi-timescale panels (Figure 10) are active-session medians of those per-session rates, with bootstrap 95% CIs, not a single pooled rate per bin. Here, an active session is one with a positive pulse rate in that bin; sessions with no recording effort or zero pulses are excluded from the median. Yearly panels therefore report session-median rates in Hz, so partial calendar years 2023 and 2026 are not scaled by the fraction of the year recorded; those years remain samples of the months that were actually logged.

5.4.4 Behavioral and environmental correlations

Temperature and conductivity from tank water logs are correlated with daily and monthly rates. Feeding windows ($\pm$5 min around logged feeding times) are contrasted with baseline minutes ($>$5 min from feeding) using mean pulse rates and a two-sided Mann–Whitney U test on the minute-level rates. Peri-feeding trajectories are computed as well. Mean feeding clock time is the circular mean of the first logged feeding-related timestamp per session, with a bootstrap 95% CI.

5.4.5 Position along the electrode line

For each accepted pulse, head position along the Berlin line is taken as the electrode with the largest positive peak amplitude in the multi-channel snippet (25 cm bins). Channel 0, closest to the logger, lay in the bright pool; later electrodes extend toward the dark pool. The bright versus dark boundary used for occupancy analyses sits at about 2.5 m along this line. Occupancy histograms and mean position are computed on the same timescales as the activity rates, including bright versus dark occupancy relative to that boundary. Collapsed occupancy rates are electrode pulse counts divided by total recording seconds (the sum of day of year recording time bins).

5.4.6 Code availability

Hardware repositories are linked above. Peak detection and video sync live in eeltracker (https://github.com/weygoldt/eeltracker) and videosync (https://github.com/weygoldt/videosync). The Berlin analysis pipeline used for this thesis (special pulse classification through position plots, matching `run_analysis.sh`) is at https://github.com/saraheisele/mscthesis.

6 Results

All results below come from continuous multi-electrode recordings of two adult captive *Electrophorus voltai* (one male, one female) in the Museum für Naturkunde Berlin tank. Most sessions used a 16-channel electrode line (25 cm spacing, 3.75 m total length) and span November 2023 to March 2026 (165 recording days; Table 1). Figure 3 shows a short excerpt of one electrode channel with both low- and high-amplitude EODs in the same window.

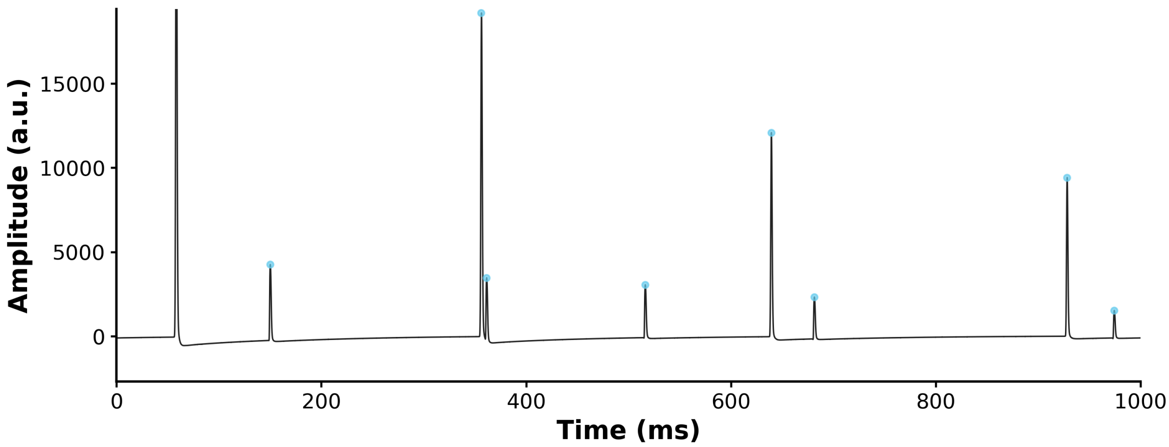


Figure 3: One-second excerpt from a Berlin tank recording (single electrode channel). Low- and high-amplitude discharges appear in the same window. Both enter the downstream counts and shape classes; there is no amplitude or volley filter.

6.1 Special pulse shapes

Waveform shape can reveal signalling diversity that discharge rate and amplitude alone do not capture. We therefore examined the shapes of all detected EODs to test whether the long-term Berlin recordings contain structured waveform variation beyond a single stereotyped pulse form. We detected $n = 13{,}465{,}438$ EODs in the processed Berlin dataset. Throughout this thesis, "all pulses", "all EODs", and related totals mean every accepted detection, including low-voltage search pulses and high-voltage discharges (shocks/volleys); events are not split or excluded by amplitude or behavioural context. The PCA plus random forest (RF) pipeline assigns each accepted EOD to one of three shape labels—normal, wide, or double (Figure 4). Median prototypes were calculated from $20{,}000$ random samples of each label to summarise the morphological extremes of the labelled set (Figure 5). Normal pulses make up the biggest share with 64.0% and a total of $n = 8{,}623{,}379$ pulses. The normal pulse is a single, roughly symmetric peak with median half width of 1.58 ms.

Wide pulses make up about a third of the dataset (33.5%, $n = 4{,}514{,}438$). A typical wide pulse is a single peak with a slower decay and median half width of 2.19 ms. The decaying part of a wide pulse is often characterized by a little "bump" or shoulder, which is not present in the normal pulse. Therefore wide pulses are also less symmetrical compared to normal pulses. Double pulses are quite rare and only make up 2.4% ($n = 327{,}621$) of all recorded pulses. Average double pulses have two peaks of similar height separated by a shallow trough. In the labelling process however, the goal was to find as many double pulses as possible to improve classifier performance. To this end, pulses that did not have the typical morphology of two clearly elevated peaks but a more or less pronounced shoulder were also labelled as double pulses. Peak to peak separation on the median double is 1.00 ms. Median half width is 2.40 ms when the waveform is treated as one event. This makes doubles the longest of the three classes, in median almost 1 ms longer than normal pulses. But the half width is not a sufficient label by itself because the classes overlap in width.

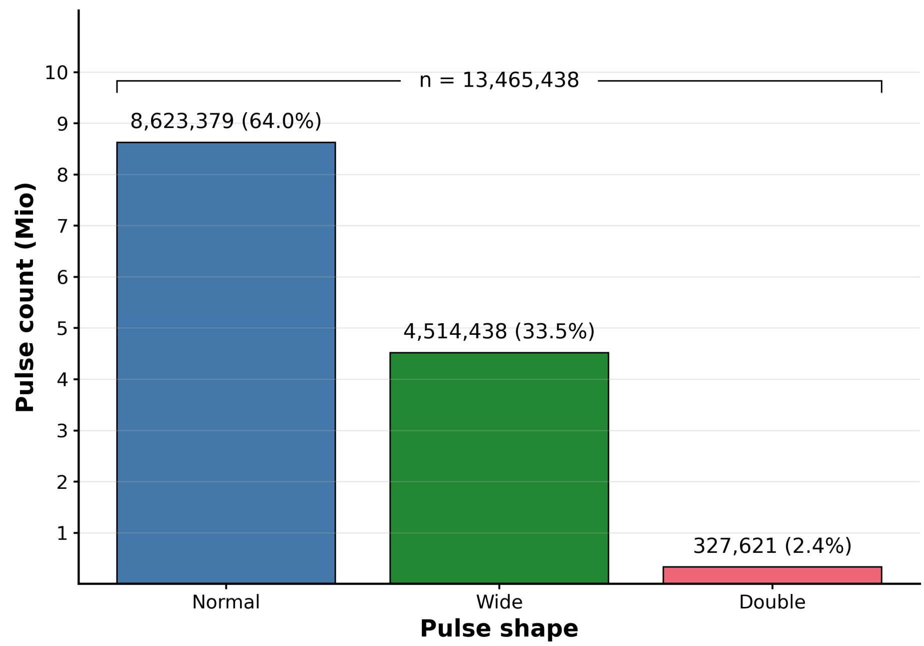


Figure 4: Number of EODs in each waveform cluster. Counts and percentages are displayed on the bars. The total includes all accepted detections regardless of amplitude (low- and high-voltage discharges). Most discharges are normal, about a third are wide, and doubles are uncommon.

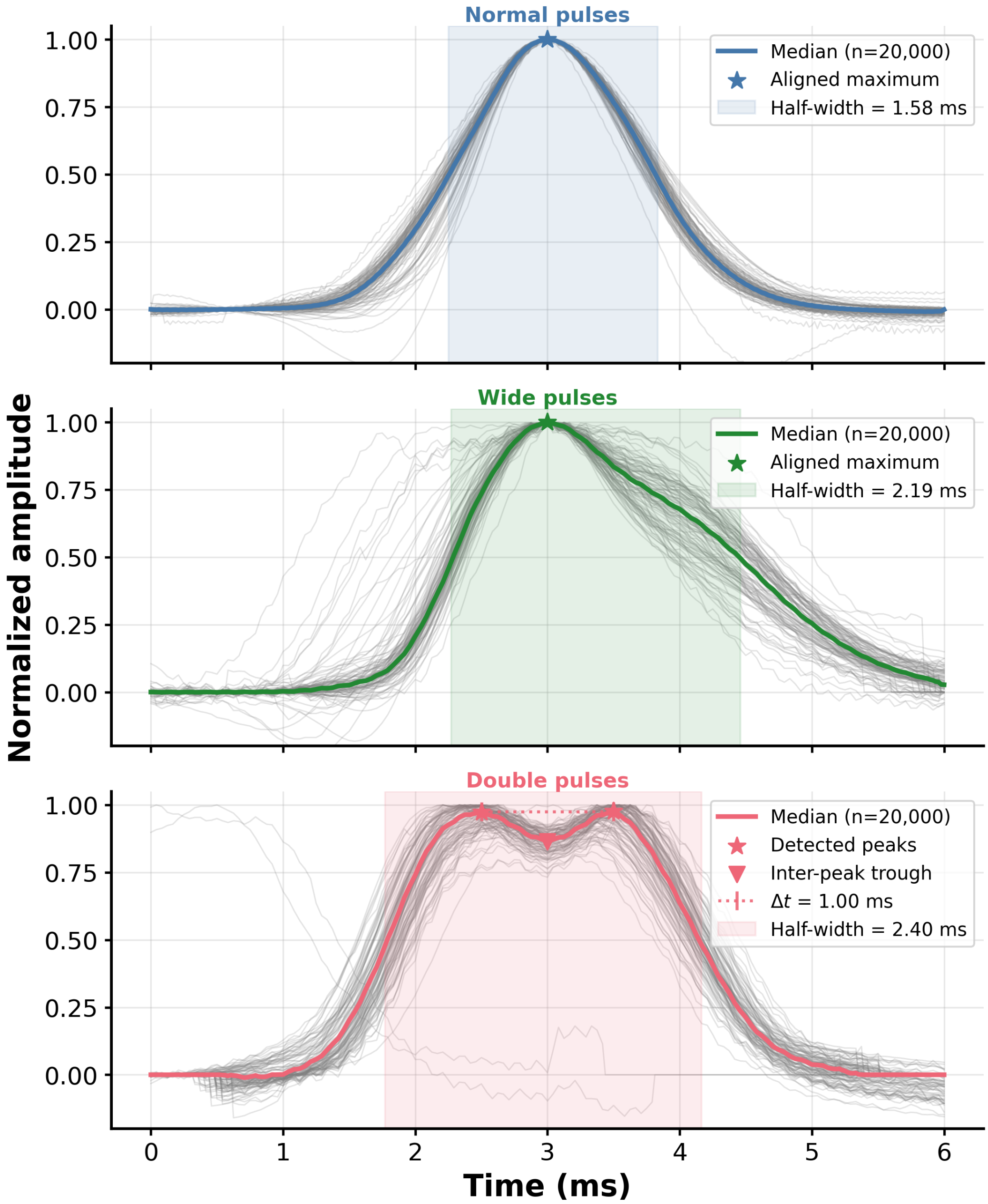


Figure 5: Median EOD waveforms for the three classes. The thick coloured trace is the class median obtained from an $n = 20{,}000$ subsample. The thin grey pulses are 100 example pulses per class. The stars mark where the waveforms were aligned (peak for normal and wide, trough between peaks for double). Normals are short and single peaked, wides stretch out with a slower falling edge, and doubles carry two peaks about 1 ms apart.

Half width distributions show graded shifts among the three labels rather than cleanly separated modes (Figure 6). Normal pulses aggregate near ∼1.5 to 1.7 ms (panel median 1.56 ms). Wide pulses shift mass a little longer (median 1.94 ms, mode near ∼2 ms). Doubles show least variability in half width distribution. The double pulses form a tight peak near 2.3 to 2.5 ms (median 2.38 ms).

The 0.7 ms half-width threshold was applied to drop near-zero and otherwise spurious widths that mostly reflect noise or misclassified events rather than genuine EOD morphology. Importantly, the left-panel n values are not the full class counts from Figure 4: they count only pulses with a measurable half width of at least 0.7 ms. Most double waveforms do not yield a finite half-width estimate (only $n = 24{,}031$ of $327{,}621$ doubles), so their legend n is much smaller than the class total. Normal and wide share substantial support around 1.5 to 2 ms, and the labelled distributions grade into one another, so width alone does not delineate the classes. The pooled half width histogram without class labels is dominated by a mode near ∼1.6 ms with a long shoulder toward 2 to 2.5 ms (Figure 6, right), consistent with overlapping, continuously varying widths rather than three discrete peaks. In the pooled histogram the peak of double pulses at about 2.5 ms is not as pronounced as in the individual class histograms. This is because the normal and wide pulses occur much more frequently compared to the doubles, which causes their distributions to dominate the overall shape of the pooled histogram. The smaller number of double pulses results in a less distinct peak in the combined distribution. Although we can't observe a clear bimodal distribution in the pooled histogram, the presence of the shoulder indicates that there is a subset of pulses with longer half widths.

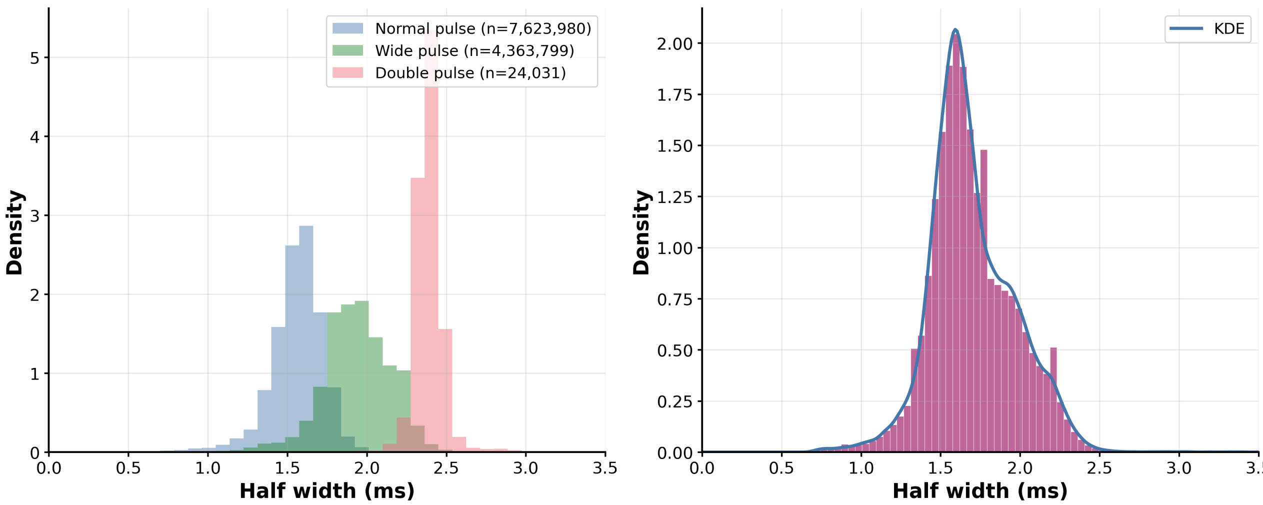


Figure 6: Half-width distributions for the three waveform classes. Left: class-wise densities (normal blue, wide green, double red); legend n is pulses with measurable half width ≥ 0.7 ms (not the full RF class totals). Right: all classes pooled as a histogram with KDE overlay. Pulses shorter than 0.7 ms are excluded to suppress noise / misclassification artefacts; the x-axis stops at 3.5 ms. Normal and wide share support around 1.5 to 2 ms, while doubles form a narrow ridge near 2.4 ms. The pooled mode sits near 1.6 ms with a shoulder toward 2 to 2.5 ms.

A PCA of the classified waveforms shows a similar picture of overlap rather than three cleanly separated clusters (Figure 7). PC1 and PC2 together explain 84.8% of variance in the plotted sample (63.3% on PC1 and 21.5% on PC2). Doubles form a tightly packed cloud, while normals spread more widely along PC2. Wide pulses sit between them and partially overlap the double region; their scatter is closer to that of normals than of doubles. In the PC3/PC4 views, doubles themselves split into two clumps along PC1. Taken together with the half-width distributions, the PCA cloud is consistent with a morphological continuum in which labelled classes occupy overlapping regions rather than discrete islands.

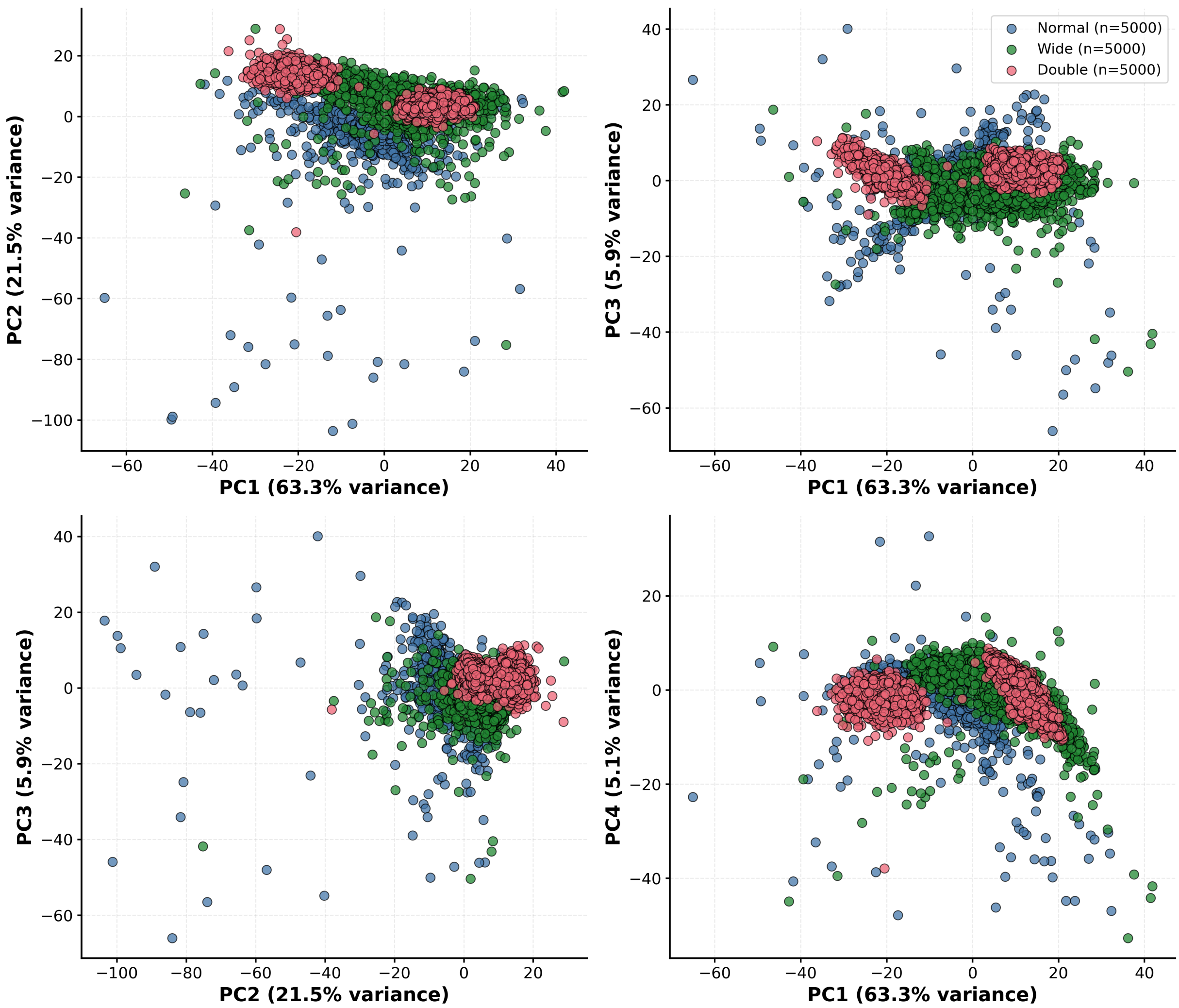


Figure 7: RF labelled pulses in PCA space, 5000 examples per class. Blue points represent normal pulses, green wide pulses, and red double pulses. The four panels are different PC pairings. Doubles form two dense clouds; normals show higher variability, especially along PC2. Wides lie between normal and double regions and partially overlap the double cloud.

A long standing question since we found evidence of double pulses was how they are produced physiologically. Aligning two of the median normal pulses with the peaks of the median double pulse sparked the hypothesis that a double peaked pulse could be the result of two normal pulses fired in quick succession. To test whether a double pulse is simply two normal pulses fired within 1 ms, the median double was modelled as the sum of two median normal templates (Section 5.4.2). When two median normal pulses are aligned to the two peaks of the median double, they already resemble the double envelope with surprising precision (Figure 8). Several nested models then fit amplitudes a_1 and a_2, peak times t_1 and t_2 (hence Δt), and component width scales to the median double (Table 2; Figure 9). Constraints on Δt and the width scales were loosened in each subsequent model; amplitudes remained free throughout. Model A starts with native widths ($w = 1$) and free Δt ($R^2 = 0.978$; Figure 9). The high R^2 in Model A is misleading: models A–D do not recover the two-peak envelope. The characteristic trough appears only when the two normal components are allowed to widen. Model E reaches $R^2 = 0.999$ with free amplitudes ($a_1 = 0.88$, $a_2 = 0.94$), free $\Delta t = 1.60$ ms, and a shared width scale $w = 1.29$. Relative to the median normal half width of 1.58 ms, that scale corresponds to fitted component half widths of 2.04 ms each—wider than a native normal, not narrower. Model F, with independent width scales, fits equally well ($R^2 = 1.000$; component half widths 1.89 ms and 2.21 ms) but does not change the picture, so E is the working description. The double is therefore not just a normal pulse repeated about 1 ms later: recovering the trough requires broader components than the normal prototype.

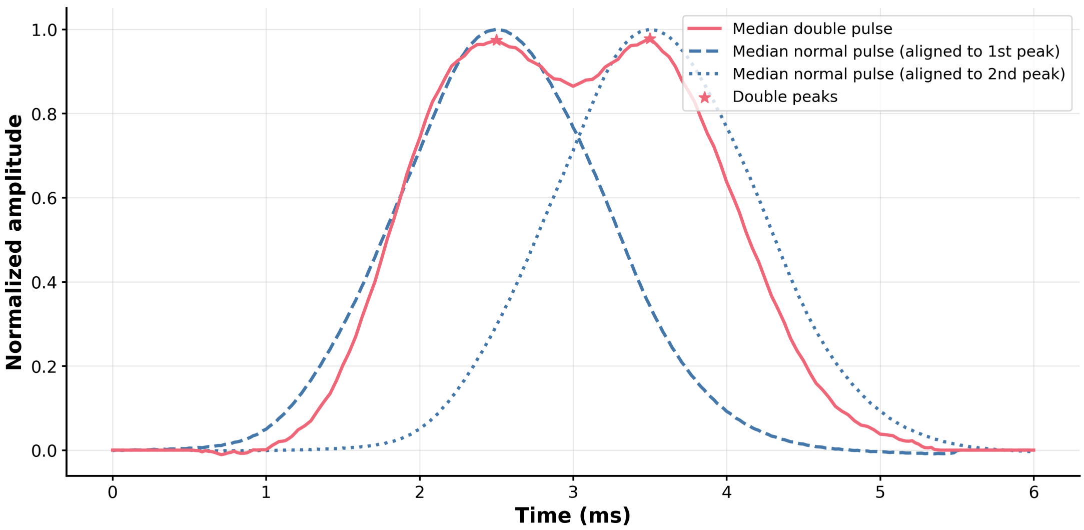


Figure 8: Two median normal pulses aligned to the peaks of the median double pulse. The solid red trace is the median double, with red stars on its peaks. The dashed and dotted blue curves are the median normal, shifted along the x-axis so each sits under one of those peaks. The trough lands where the two normals cross. A pure short doublet of unmodified normals already looks similar, but the quantitative fits (Figure 9) show that matching the trough requires broader components.

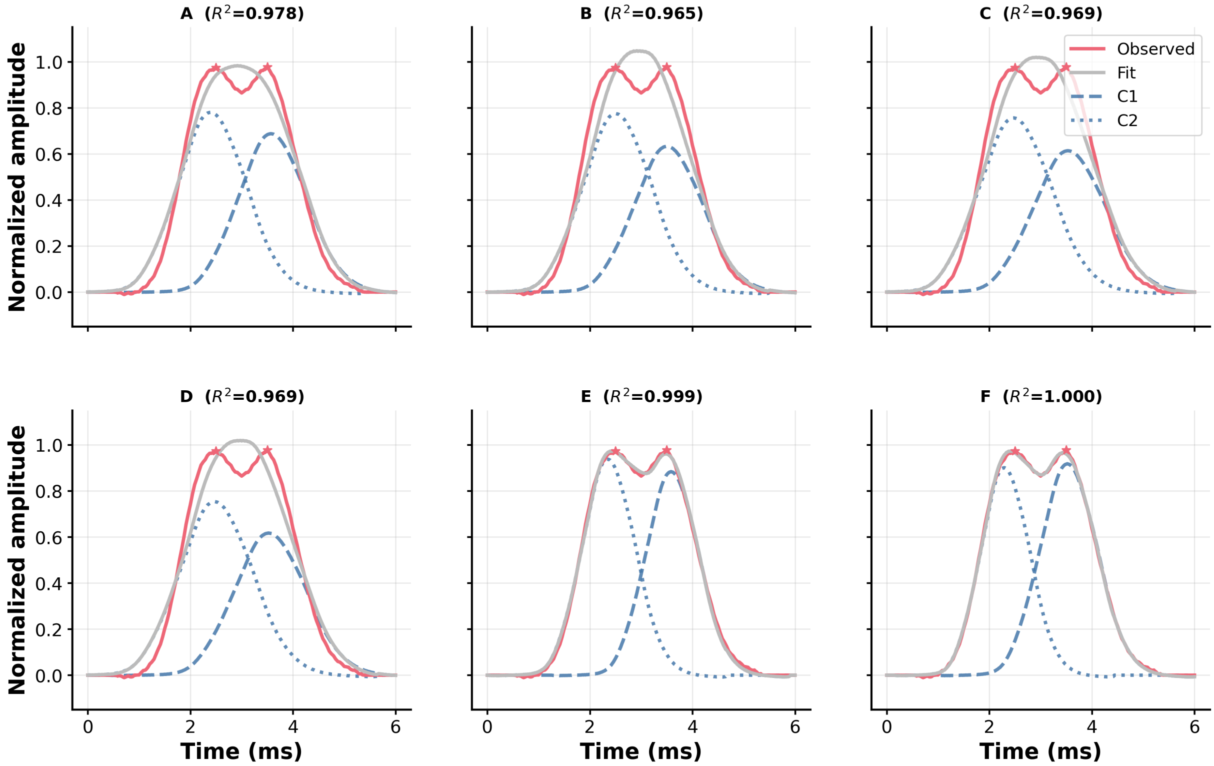


Figure 9: Six ways to model the median double pulse from two median normal components (constraints in Table 2). Each panel is labelled by model letter and R^2 and shows the median double (red, stars on the peaks), the model sum (grey), and the two normal components C1 and C2 (dashed and dotted blue). From A to F the constraints on Δt and w get looser ($w = 1$ is the native normal width; $w > 1$ widens). A already reaches $R^2 = 0.978$ with native widths. E and F let widths float and push R^2 near 1, with fitted Δt drifting away from the observed 1 ms peak spacing and with components broader than the native normal. The characteristic trough of the double pulse shape is recovered only by models E and F.

6.2 Activity across timescales

Pulse rates normalized for recording effort are shown separately for all EODs, wide pulses, and double pulses at four timescales in Figure 10. Across the full dataset the effort-normalized mean rates are 1.21 Hz across all EODs. The special pulse shape rates are 0.38 Hz for wide, and 0.018 Hz for double pulses. Medians in the panels are active-session medians with bootstrap 95% confidence interval (CI) shading: only sessions with a positive pulse rate in that bin enter the median, so empty or silent sessions are excluded.

On the 24 h scale (Figure 10, top row), we can observe th circadian rhythm. The all pulse rate sits near ∼0.6 Hz through the night (00:00 to 04:00), rises after about 05:00, and peaks near 08:00 to 09:00 at roughly 1.55 Hz. After that morning peak the median falls to about 1.1 Hz by 10:00 and ranges between roughly 0.75 and 1.0 Hz for the rest of the day. Double and wide medians stay near zero across the clock, with wide pulses showing the same pulse rate elevation in the morning as the all pulse curve. That morning maximum lines up with feeding related activity (Section 6.3).

The calendar month scale follows a similar pattern (Figure 10, second row). All pulse medians sit around 0.6 to 1.0 Hz for most of the year and there is one clear peak in November ($\sim$1.7 Hz) that decreases in December and returns to baseline in January. There are two noticable peaks in the confidence band at March and May rate increase in March. As seen in the 24h scale, wide pulses track that November bump from all pulses. They are near zero from February through September, rise in October, and reach about 0.6 Hz in November before returning to baseline in January. Double pulses stay at a near-zero baseline of roughly 0.018 Hz all year without any noticeable peaks or patterns.

Months since recording start (Figure 10, third row) adds more information to explain the increased late year pulse rates in the calendar month view. There is one peak in November 2025 ($\sim$2.4 Hz), indicating that the November bump is not seasonal but can be attributed to specific behavior in 11/2025. Beyond that, early 2024 carried the highest all pulse medians, with peaks near $\sim$6 Hz in April 2024 and $\sim$5.6 Hz in June 2024. It is important to add here that the CIs are wide in those months. Also, these peaks are not reflected in the calendar month view although the November 2025 peak is. It is likely that they are driven by a few sessions with high pulse rates rather than a representative increase in activity across sessions. After June 2024 the all pulse median falls sharply: July 2024 is a single overnight session near $\sim$0.9 Hz, and August to October 2024 remain near $\sim$0.7 to 1.8 Hz. Gaps appear where no sessions contribute. This is the case for January 2024 and February 2025. Wide pulses are almost absent until the November 2025 spike ($\sim$1.4 Hz). Double pulses stay low, with a small rise towards the end of the recordings on the order of 0.5 Hz into early 2026.

Yearly bins (Figure 10, bottom row) show a steady drop in all pulse median rate, from about 3.5 Hz in 2023 through 1.7 Hz (2024) and 1.0 Hz (2025) to about 0.5 Hz in 2026. The bootstrap band also narrows over those years. Dual-line weighting (Methods) is already applied in these session medians, so the drop is not an uncorrected double count from the second electrode line. 2023 and 2026 are still partial calendar years, so those two points sample only the months that were recorded. Double and wide yearly medians remain near $\sim$0.1 Hz and do not follow that long-term decline.

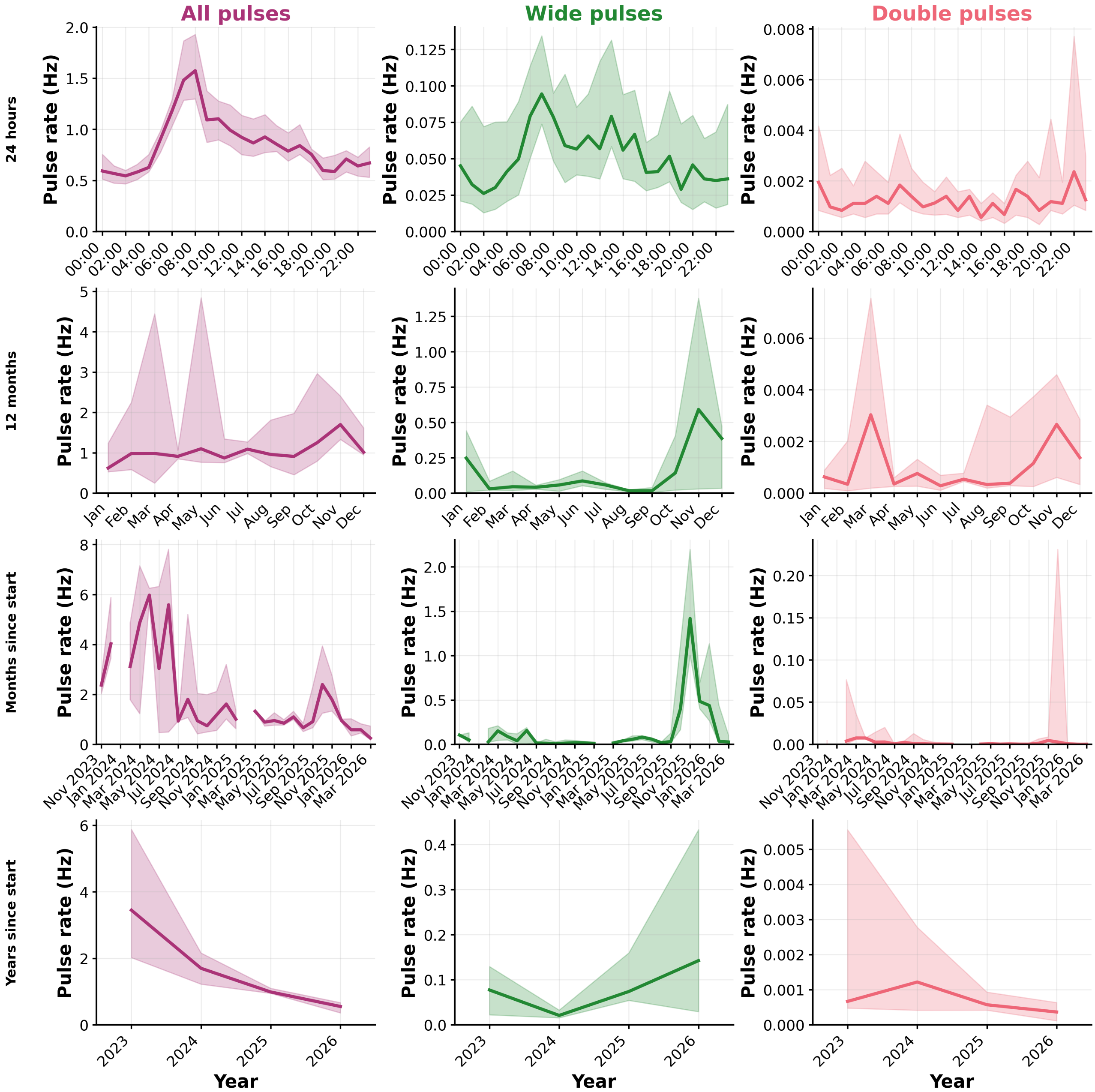


Figure 10: Pulse rate across different timescales, shown separately for all pulses (left), wide pulses (middle), and double pulses (right). Thick traces are active session medians; washed-out bands show bootstrap 95% CIs. Rows from top to bottom: 24 h (hourly bins), calendar months, months since recording start, and yearly bins from 2023 to 2026.

6.3 Environmental and feeding correlations

Monthly means of pulse rate against tank temperature and conductivity show little structure in the measured ranges (Figure 11; $n = 13$ months with overlapping sensor logs). Temperature spans only about 25.4 to 26.8 °C. Conductivity roughly spans from 180 to 350 μS cm^{-1}. Pearson tests on the plotted linear fits are flat for all pulses (temperature: $r = -0.06$, $p = 0.84$; conductivity: $r = 0.01$, $p = 0.96$) and for wide pulses (temperature: $r = -0.14$, $p = 0.65$; conductivity: $r = -0.38$, $p = 0.20$). Double-pulse rate vs temperature is the only significant linear association ($r = 0.62$, $p = 0.023$), but the matching Spearman rank correlation is not ($\rho = -0.07$, $p = 0.81$). So that correlation is not robust to rank order and likely reflects a few leverage points in a small sample. Double vs conductivity is likewise non-significant ($r = 0.23$, $p = 0.44$). Wide and double rates sit much lower than the all-pulse scatter. With only $n = 13$ months, the informative value of this analysis is limited meaning that existing correlations could be missed.

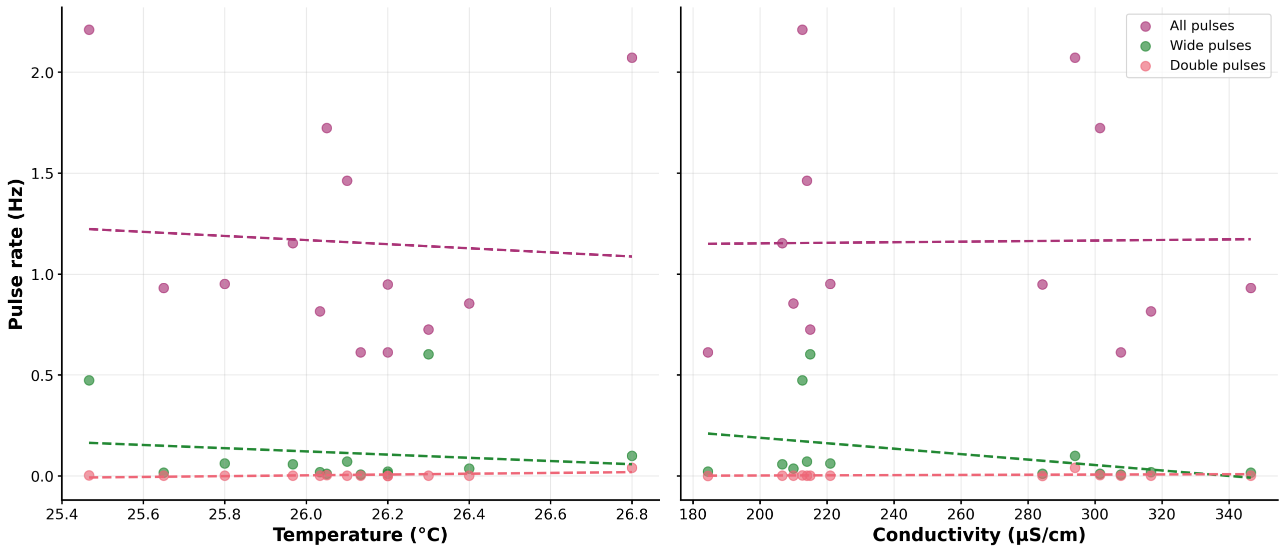


Figure 11: Environmental correlations of pulse rate in Hz with tank temperature in °C (left) and conductivity in $\mu S\,cm^{-1}$ (right). Each cloud is a pulse category (purple - all pulses, red - double, green - wide), with a dashed line of the same colour indicating the linear fit ($n = 13$ months). The only significant Pearson correlation is double vs temperature ($p = 0.023$). In a Spearman rank correlation all three traces are not significant. Special-shape rates sit well below the all-pulse scatter.

Correlations of activity and feeding times show a much clearer picture. Feeding windows were defined as ±5 min around the logged feeding time, and baseline windows as >5 min away from feeding. Mean pulse rate for all pulses is 1.34 Hz during non-feeding times, versus a 3.8-times increase to 5.03 Hz in feeding windows (Figure 12; two-sided Mann–Whitney U, $p < 0.001$). Wide pulses rise from 0.45 Hz to 0.64 Hz (1.4-times; $p < 0.001$). Double-pulse window means stay near 0.03 Hz (0.033 Hz baseline vs 0.030 Hz in feeding windows), so the mean contrast does not show a net increase. Still, the minute-level distributions differ significantly($p < 0.001$) because doubles occupy a larger fraction of feeding minutes. On its own peri-event scale (Figure 13), the double-pulse rate shows a brief peak near 0.05 Hz at the logged feeding time from a near-zero pre-feeding baseline. Peri feeding trajectories for all pulses show the rise starting about 2 min before the logged feeding time, peaking near 9 Hz at time zero, then decaying over the next 10 min. Wide pulses follow the same shape at lower amplitude (peak near 1.1 Hz). Mean feeding clock time is 08:34 (bootstrap 95% CI 08:14–08:57; $n = 127$ sessions). That morning rate peak in Figure 10 coincides with feeding times.

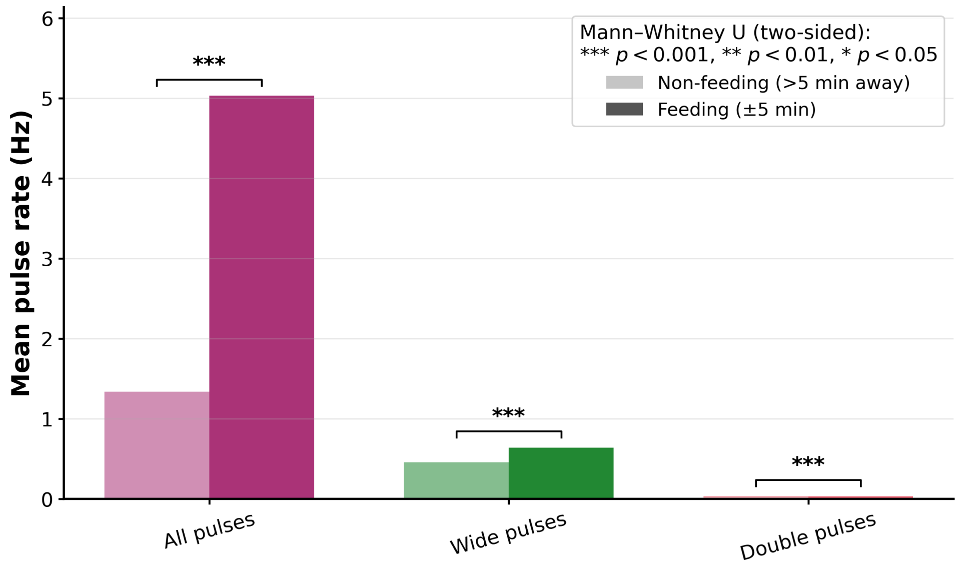


Figure 12: Mean pulse rate in feeding windows (±5 min) next to the non feeding baseline (>5 min away). Purple, green, and red are all, wide, and double pulses. Within each hue the darker bar is feeding and the lighter one is baseline. Brackets mark two-sided Mann–Whitney U tests on minute-level rates ($***$: $p < 0.001$). Overall activity and wide pulses increase during feeding; doubles stay near the x-axis on this shared mean scale but still differ by the rank test.

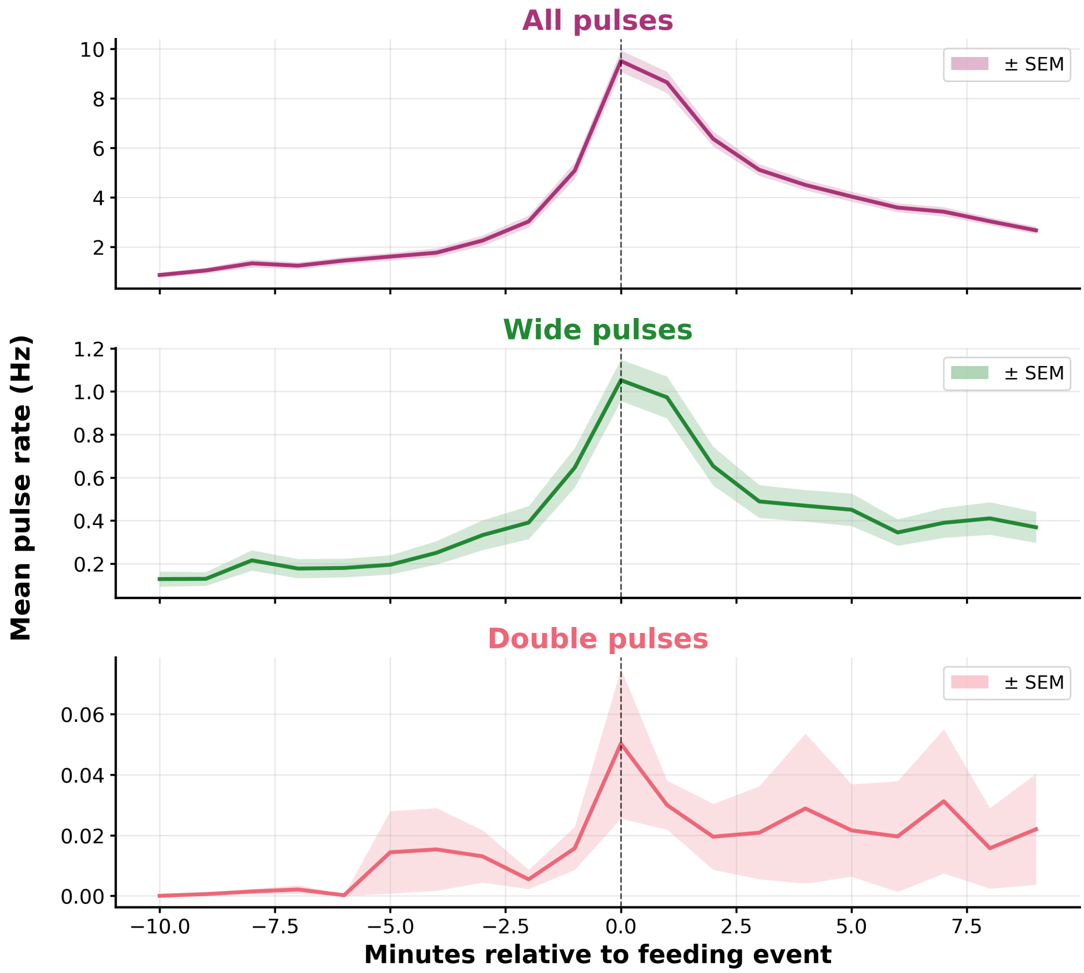


Figure 13: Minute by minute rate around feeding, one panel each for all pulses, wide pulses, and double pulses, each with its own y-axis. Coloured traces are means across matched feeding events (purple - all pulses, green - wide, red - double). The matching washed out band is $\pm$ SEM over those events (`scipy.stats.sem` in `correlate_activity_with_feeding.py`). The dashed vertical line is the logged feeding time that each trace was aligned to. Overall rate starts climbing about two minutes early, hits roughly 9 Hz at time zero, then decays until the end of the time window at $+10$ minutes. Wides follow the same shape at lower amplitude (peak near 1.1 Hz). On the double-pulse panel a short peak near 0.05 Hz appears at time zero.

6.4 Position estimation

Head position was taken as the electrode with the largest positive peak in each multi channel snippet. Collapsed over all hourly bins, occupancy is strongly end biased (Figure 14). Pulse rate peaks at electrode 0 (bright end, $\sim$0.33 Hz) and electrode 15 (dark end, $\sim$0.36 Hz). Mid bright electrodes (1 to 9) stay low. Dark side electrodes 10 to 14 carry more mid line activity than the bright mid line. The animals park at tank ends and use the dark half more while moving between ends.

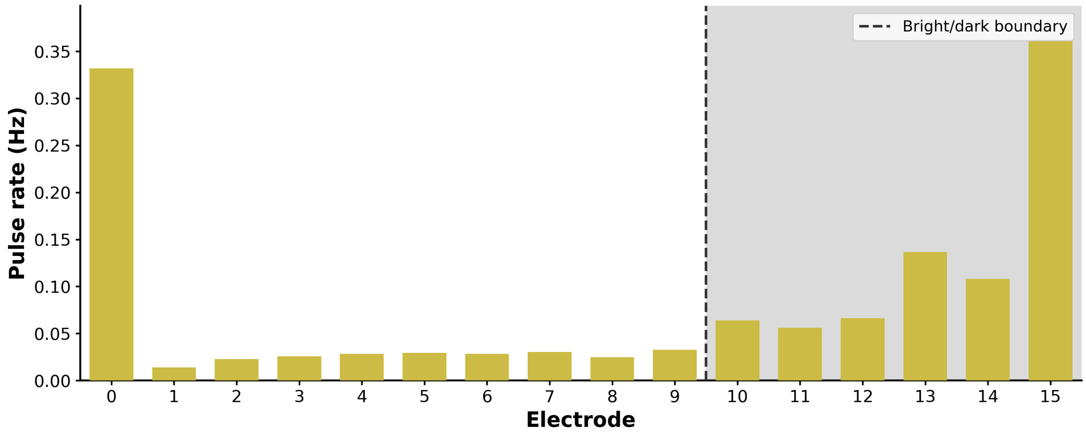


Figure 14: Where along the 16 electrode line the pulses land when hourly bins are collapsed. Yellow bars give pulse rate per electrode. The dashed vertical line is the bright/dark cut at 2.5 m, and the light grey wash marks the dark half (electrodes 10 to 15). Animals spend most time at the two ends (0 and 15), and the dark mid electrodes stay busier than the bright mid stretch.

Session wise median head position across timescales is shown in Figure 15. On the 24 h scale the median stays mostly on the bright side of the 2.5 m boundary, dipping toward the front (lower meters) around 07:00 to 08:00 and sitting closer to the dark boundary at other hours. The median alone understates how often the upper percentile band crosses into the dark half. Calendar month and months since start show slow drifts of preferred position, including a late 2025 push toward the dark boundary.

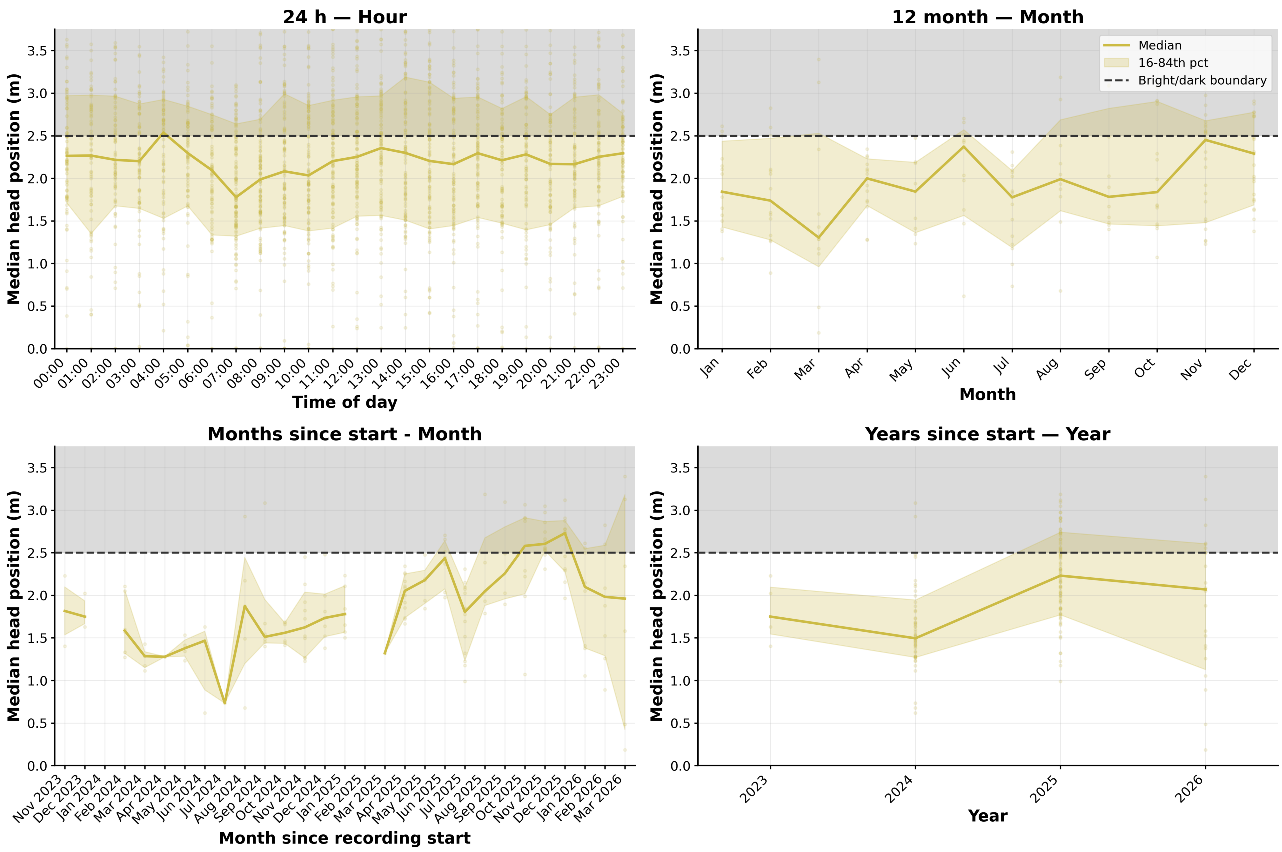
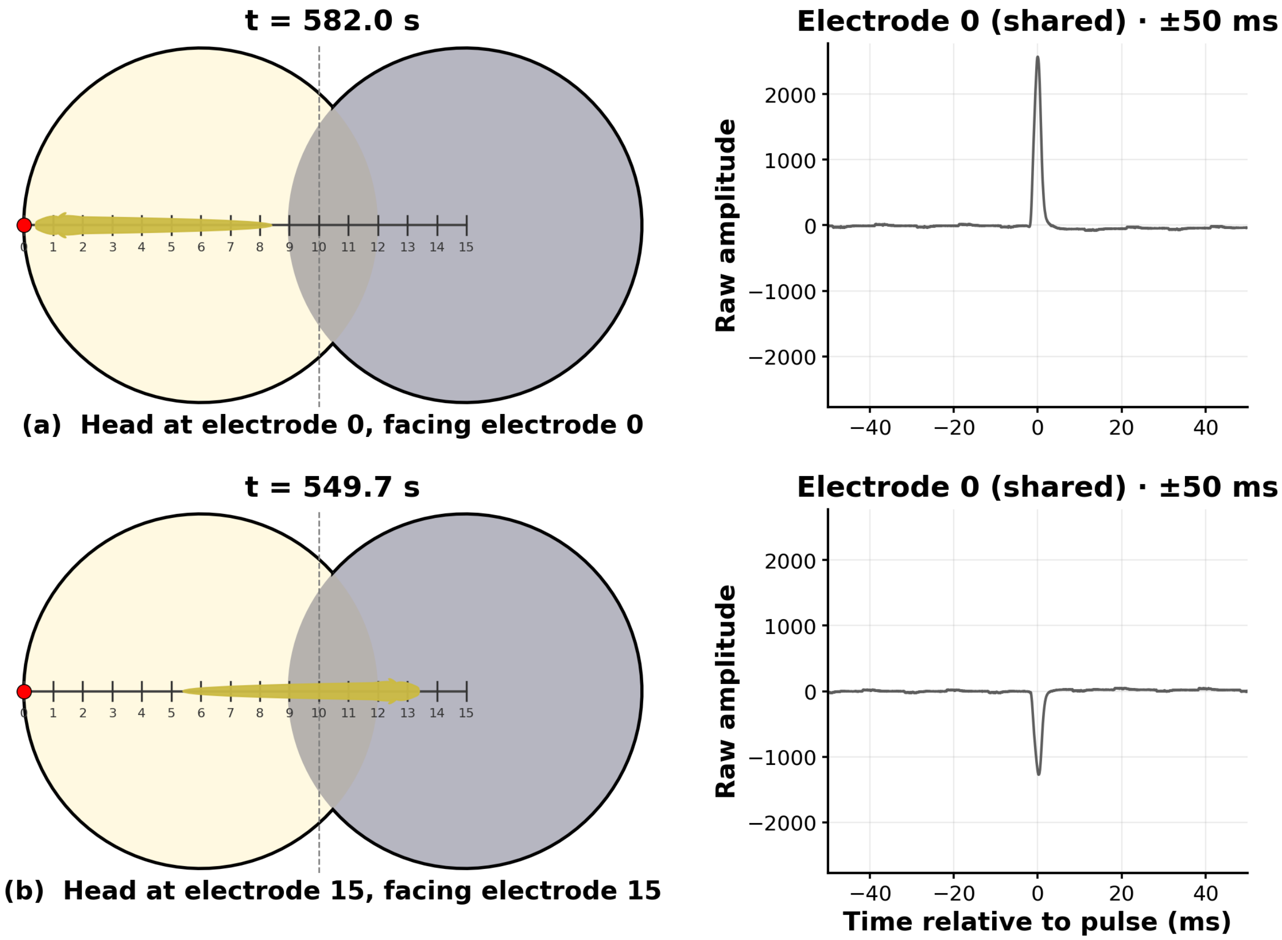


Figure 15: Head position through the same four timescales as the activity figure. The yellow line is the session wise median, the light yellow band the 16th to 84th percentile, and the faint yellow dots the session values themselves. Everything above the dashed line (grey wash) is the dark half of the tank past 2.5 m. Medians usually sit on the bright side, slide toward the front mid morning, and wander over months, including a late 2025 stretch nearer the dark boundary.

Figure 16 shows selected frames from a short turnaround scene in one recording. An eel icon is plotted to move along the electrode line with the head position equal to the strongest positive electrode. The printed figure is a two-panel keyframe strip; the full short animation (GIF) is available via the QR code.



(a) Head at electrode 0, facing electrode 0

(b) Head at electrode 15, facing electrode 15

Scan the QR code for the head-position animation (GIF). Direct GIF: https://raw.githubusercontent.com/saraheisele/mscthesis/master/docs/latex_thesis/figures/position_estimation/eel_position_animation.gif.

Figure 16: Two frames from one Berlin recording (dark-area logger chunk `eellogger02-20260128T163148.wav`). Panel (a): head at electrode 0, facing the bright end; the raw trace shows electrode 0 with a large positive EOD. Panel (b): head at electrode 15, facing the dark end; the same electrode 0 trace is polarity-flipped and much smaller. Head placement uses the electrode with the strongest positive peak. In this printed figure both raw panels deliberately share fixed electrode 0 so amplitude and polarity can be compared on one channel; in the GIF animation the right-hand raw panel instead follows each pulse's head electrode (the channel with the strongest positive peak at that time). The bright (yellow) and dark (grey) tank halves are indicated by the overlapping circles, with the dashed line at the bright/dark boundary. The QR code opens the matching GIF supplement.

7 Discussion

This thesis draws on a multi-year record of electric organ discharges (EODs) from captive *Electrophorus voltai*. The analysis covers pulse shapes, pulse rates across several timescales, coarse space use along an electrode line, and the response of pulse rate to feeding. The dataset contains more than thirteen million accepted EODs and spans approximately 14 TiB of raw recordings. A big share of this work also includes the hardware development and field work that made these recordings possible. In that sense, the thesis is as much a methodological contribution as an ethological one. The main ethological value of the dataset is that its duration makes it possible to identify changes in EOD activity that would be difficult to distinguish from short-term variation in a conventional behavioral observation.

Across the Berlin dataset, the classifier assigns accepted EODs to normal, wide, and double labels. These labels are useful working categories for summarising waveform variation, but neither the half-width distributions nor the Robust PCA support three cleanly separated morphological islands. Normal and wide pulses overlap substantially around 1.5 to 2 ms, while doubles form a narrower ridge at around 2.4 ms; in PCA space, wides sit between normals and doubles and partially merge with the double cloud. The overall picture is closer to a continuum of waveform morphology, across which pulses can vary in structural properties such as duration, shoulder strength, and relative peak timing. That continuum makes hard boundaries between classes difficult to justify and means that any discrete label scheme remains an approximation. At the same time, the extremes of this continuum remain striking: short symmetric single peaks, elongated single peaks with a shoulder, and clear two-peak envelopes differ enough that describing them is a necessary first step toward asking why EOD shape is so diverse and how it is manipulated. Candidate explanations include voluntary or involuntary behavioural state as well as external factors such as stress or physical conditions; the present data do not yet distinguish among them.

Previous work on *Electrophorus* has mainly distinguished discharge forms by amplitude and behavioural context. For example low-amplitude search pulses, high-amplitude hunting volleys (Catania 2019), and a third form that differs primarily in amplitude rather than in waveform shape (Xu et al. 2021). The present analysis keeps low- and high-voltage discharges in one accepted set and then groups them by waveform morphology, independently of amplitude or of whether they occur in search activity or in high-rate volleys. Rate, feeding, and position analyses use that same inclusive set. In that sense, the Berlin dataset provides, to our knowledge, the first large-scale, long-term characterization of this shape variation in *Electrophorus*. The labels describe waveform shape and do not, by themselves, identify behavioural acts. This distinction is important throughout the interpretation of the results.

In the higher PC views, double pulses themselves split into two clumps along PC1. This may reflect more than one double morphology, for example differences in relative peak height or inter-peak timing, rather than a single stereotyped waveform.

The median double pulse can also be approximated by two normal-like components separated by about one millisecond. When two median normal pulses are aligned with the two peaks of the median double, their overlap produces a trough close to the one observed in the double pulse. A two-template fit recovers that envelope only when the components are allowed to widen relative to the native normal (median half width 1.58 ms to fitted 2.04 ms in model E), so the double is not simply a normal pulse repeated in quick succession. The critical result is that the waveform is consistent with a short doublet of broadened normal-like components, not that the double is generated by two unmodified normals.

These morphological findings should also be distinguished from previously described high-voltage doublets. High voltage doublets have been described during prey capture, where they can trigger prey movement before the suction strike (Catania 2014). However, the double peaked pulses described in this work differ strongly from those doublets in that they have been observed in varying signal strengths, not only in high-voltage volleys. In addition, doublets are described as two separate pulses while the double pulses in this work are described as a single pulse with two peaks. The present wide pulses are more closely related to the single-peaked waveform.

One possible physiological route to a double waveform would involve different contributions from the electric organs. A single command from a pacemaker structure might produce one component, while a short burst recruits an additional organ and generates a second component on the electrode about a millisecond later. This is speculative. Work on weakly electric fish shows that ion channel kinetics can affect electrocyte spike duration and therefore EOD pulse width McAnelly and Zakon (2000), which makes a physiological contribution to waveform fine structure plausible, but it does not provide a direct explanation for the double pulses observed here. The membrane physiology described for electric eels much earlier remains relevant background for the same reason (Keynes and Martins-Ferreira 1953). Possible behavioural associations of these morphological extremes are taken up below in the feeding analysis and in the late-2025 activity pattern.

7.1 Feeding-related activity

The feeding analysis provides the clearest behavioural association in the dataset. Mean all-pulse rate in the ± 5 min window around logged feeding events increases from approximately 1.34 Hz outside feeding windows to 5.03 Hz during feeding. The peri-feeding analysis shows that the increase starts approximately two minutes before the logged feeding time, reaches approximately 9 Hz around the feeding event, and then declines over the following ten minutes. This pattern also helps explain the morning maximum in the 24 h activity plot. The timing is consistent with the animals increasing their pulse rate before food arrives, which could reflect anticipation of the feeding schedule or responses to cues associated with the keeper and feeding routine.

Wide pulses show a similar response, increasing from approximately 0.45 Hz to 0.64 Hz during feeding windows. Their peri-feeding trajectory also follows the overall increase in pulse rate, although at a lower amplitude. This association suggests that wide pulses are involved in the elevated EOD activity surrounding feeding. It is tempting to relate this finding to hunting, since electric eels use their electric system extensively during prey detection and capture (Catania 2014; Catania 2015a; Catania 2015b; Catania 2019). However, the present recordings were not linked to individual prey encounters or predatory strikes. The feeding response therefore provides evidence for an association between wide pulses and feeding-related activity, but it does not establish that wide pulses are hunting signals. That limitation is unsurprising here: the eels were fed dead fish or fish pieces rather than live prey, so a classic live-prey hunting signature was not expected.

Across the three shape classes, the peri-feeding trajectories share a similar relation to the feeding event, although at very different absolute rates. Mean double-pulse rates remain close to baseline in the feeding windows, yet the minute-level distributions differ significantly and a short increase is visible around the logged feeding time in the peri-event analysis. Rather than isolating double pulses as a feeding- or hunting-specific signal, this pattern is more consistent with waveforms that change only slowly while the underlying electrolocation function of the EOD is retained across shape classes. The morphological similarity to hunting doublets described during prey capture (Catania 2014) is therefore interesting, but similarity alone is insufficient to assign the Berlin double-pulse class to hunting behaviour. Directly video-linked recordings of feeding events would be a useful next step for resolving this question.

7.2 Temporal patterns and the late-2025 increase

The activity analysis reveals several patterns on different timescales. The 24 h activity profile contains a pronounced morning increase, which coincides closely with the regular feeding schedule. On the calendar-month scale, overall and wide-pulse rates show a clear increase in the later part of the year, with the strongest wide-pulse increase occurring around November. However, the chronological analysis is important for interpreting this pattern. Much of the November increase is caused by a pronounced wide-pulse increase in November 2025 rather than by a pattern that recurs in the same calendar month across years. Wide pulses are almost absent during much of the earlier recording period and reach their clearest increase during this late 2025 period. Double pulses remain much rarer, but also show a smaller increase toward the end of the recording period.

This makes a simple seasonal explanation unlikely. If the increase were a seasonal pattern, a similar increase would be expected during the corresponding months in more than one year or with a different temporal structure. Instead, the strongest increase is restricted to one specific period late in the recording. This does not rule out environmental influences completely because the captive animals were still exposed to some environmental and husbandry cues. During the same broader late-2025 window, logged tank conductivity sat toward the lower end of the measured range (roughly 180 to 220 μS cm^{-1} in mid-to-late 2025, compared with roughly 290 to 370 μS cm^{-1} in the overlapping 2024 months). November and December 2025 themselves have little overlapping conductivity coverage, so this remains a temporal coincidence rather than a demonstrated environmental driver. However, the present data do not support interpreting the November 2025 increase simply as a recurring seasonal or rainy-season pattern.

The late-2025 increase becomes more interesting when considered together with the spatial and behavioural observations. The position analysis shows that the animals preferentially use the ends of the tank and that the dark middle electrodes are used more than the bright middle section. More importantly, the session-wise position data show a movement toward the dark boundary during late 2025. The position estimate is coarse and cannot be interpreted as a direct measurement of habitat preference, but its temporal coincidence with the increase in special-pulse activity is notable.

The dark section of the tank also contains structures such as roots that provide potential opportunities for breeding-related behaviour. This is relevant because the reproductive behaviour described for electric eels in the natural environment is closely associated with structured habitats and root masses (Assunção and Schwassmann 1995; Bastos 2020). The spatial data cannot establish that the animals were using the dark region for reproductive purposes. They only show that space use changed during approximately the same period in which overall and special-pulse activity increased.

During this period, additional observations by the animal caretaker provide anecdotal behavioural context for the quantitative recordings. The eels were described as remaining closer to one another than usual and actively seeking each other's proximity. This was considered unusual behaviour, particularly for the female. Because these observations were not collected using a standardized behavioural protocol, they cannot be treated as independent quantitative evidence. Nevertheless, their temporal coincidence with the late-2025 changes in pulse rate and space use makes them relevant when considering possible explanations for the observed activity pattern.

Taken together, the increase in overall and special-pulse rates, the movement toward the darker region of the tank, and the unusual observations of increased proximity between the animals are consistent with a change in social state. A reproductive interpretation is particularly interesting in light of the limited knowledge of electric eel courtship and reproduction. The introduction of this thesis describes reproduction as one of the least well understood aspects of *Electrophorus* biology. The late-2025 observations therefore raise the possibility that the increase in special-pulse activity was related to reproductive or social behaviour.

However, this interpretation is merely a hypothesis until there is more direct evidence. The present recordings do not identify mating, courtship, nest building, spawning, or any other specific reproductive act. The position estimate cannot distinguish individual animals reliably when several eels contribute to the recording, and the behavioural observations were anecdotal rather than systematically scored. The most appropriate conclusion is therefore that the late-2025 changes provide evidence consistent with an association between special-pulse activity and an unusual social or potentially reproductive state, rather than evidence that the special pulses are reproductive signals.

The subsequent death of the female in late December provides some additional temporal context for this interpretation. If the suspected reproductive condition, including the possibility of egg retention, can be established, this observation may be relevant to the hypothesis that the late-2025 increase was associated with reproductive state. The male died approximately four months later, which also places the two deaths within the same broader period of unusual activity. However, the cause of death and its relationship to reproduction cannot be established from the present dataset. These observations should therefore remain clearly separated from the quantitative evidence and merely provide circumstantial context for our hypothesis rather than proof of a reproductive event.

7.3 Environmental variables and long-term changes

Environmental variables provide a less clear alternative explanation for the observed activity patterns. The measured temperature and conductivity ranges did not show robust relationships with overall, wide, or double-pulse rates. The one significant Pearson correlation between double-pulse rate and temperature was not supported by the corresponding Spearman correlation and is therefore not convincing. The relatively narrow temperature range in the tank also limits the ability of this analysis to detect environmental effects. A lack of a correlation within the measured range should not be interpreted as evidence that EOD activity is generally independent of temperature. Temperature-related changes of EOD waveform have been reported in weakly electric fish, including *Brachyhypopomus* and *Gymnotus* (Ardanaz et al. 2001; Silva et al. 1999). The temporal coincidence noted above between the late-2025 special-pulse increase and a period of comparatively low tank conductivity therefore remains only a qualitative observation: it does not override the absence of a robust rate–conductivity correlation across the logged months.

A separate long-term pattern is the decrease in median all-pulse rate from approximately 3.5 Hz in 2023 to approximately 0.5 Hz in 2026. The same pattern appears as the sharp drop after June 2024 in the months-since-start panel. Both are at least partly explained by how the recordings were sampled and by hardware limits. Early 2024 highs are dominated by short morning feeding sessions with high rates, whereas from July onward longer overnight runs contribute more to the monthly medians and pull them down. Session notes from mid-July 2024 also report that the electrode line was dead from channel 9 onward, leaving only 12 channels, which could undercount pulses whenever the animals were toward the dark end of the line; processed recordings show all 16 channels again from 7 August 2024 onward. Aging, habituation, and changes in the number of animals remain possible additional factors. Dual-line pulse counts are already weighted from 25 November 2025, so the yearly decrease is not simply an effect of adding a second electrode line. The partial coverage of 2023 and 2026 also limits direct comparison between those years.

Importantly, the special pulse classes do not simply follow the same long-term decline as all pulses. This suggests that the reduction is not a uniform quieting of every discharge class.

7.4 Spatial behaviour and limitations of the position estimate

Spatial exploration of the Berlin dataset is necessarily coarse. Head position is estimated from the electrode with the largest positive peak. This estimate is most reliable when one animal dominates a short recording segment and is oriented reasonably along the electrode line. It becomes less reliable when several animals discharge simultaneously or when an animal lies parallel to the array.

Overall, the darker middle electrodes are used more often than the brighter middle section. Session-wise medians also shift across the day and across months, including a noticeable movement toward the dark boundary late in 2025. The morning movement toward this end therefore occurs alongside the feeding response. One thing to keep in mind is that the bright end is also associated with the keeper and feeding side.

The spatial scale of the tank also limits comparison with the natural environment. Improvements to the localisation procedure, including further development of the eeltracker processing chain, would make it possible to reconstruct trajectories rather than only occupancy frequencies. The FLONA data was mainly collected with electrode grids instead of line which opens up the possibility of a more complete spatial reconstruction in the natural environment.

7.5 Captivity and comparison with the natural environment

The distinction between the captive and natural environments is important for interpreting the absence of strong periodic behaviour as well as the late-2025 increase. Wild electric eels are exposed to environmental cues that are difficult to reproduce in captivity, including changes in hydrology, habitat structure, and food availability (Assunção and Schwassmann 1995; Bastos 2020).

Although the husbandry conditions were adjusted to provide potential cues for breeding, the captive animals were not exposed to the full set of environmental changes experienced in their natural habitat. A lack of a strong periodic signal in the Berlin recordings should therefore not be interpreted as evidence that electric eel behaviour itself lacks periodicity.

The Berlin dataset is particularly valuable because its controlled setting and long duration make it possible to identify waveform classes, obtain enough rare double pulses to examine their morphology, and compare pulse rates around repeated feeding events. At the same time, the captive setting cannot resolve behaviours that are unlikely to occur in the tank, including migration, natural nest building, natural prey encounters, and the group hunting behaviour reported in the field (Bastos et al. 2021). The Berlin dataset therefore provides a controlled baseline rather than a substitute for observations in the natural habitat.

The field recordings collected during the September 2025 expedition provide the most direct next step. Applying the same waveform classification and activity analysis to the FLONA and Ducke recordings would allow the special pulse classes identified in Berlin to be examined under natural conditions. This comparison could determine whether their relative rates, pulse-rate timescales, feeding-related activity, and periodic patterns are similar between captivity and the wild. It could also reveal patterns that are absent in the captive dataset because the corresponding environmental or social conditions were not present.

This comparison is important for interpreting the behavioural significance of the present findings. If similar waveform and activity patterns occur in the wild, this would strengthen the argument that they represent biologically meaningful aspects of *E. voltai* behaviour rather than responses specific to captivity. Conversely, differences between the datasets would help identify which aspects of the observed activity are associated with the captive environment.

The recording system itself is an important part of what makes these analyses possible. Pulse amplitudes measured on the grid fall within the range of tens of volts reported previously by Xu et al. (2021), providing a useful check that passive non-invasive electrodes recover the expected voltage scale. The TeeGrid, TeeRec, and Teensy amplifier system, together with the waterproofing and battery solutions developed for the field work, provide a basis for extending the same recording approach beyond the captive setting. For software availability and hardware details, see Section 5.1.

7.6 Conclusions and Outlook

The Berlin recordings provide a long-term captive EOD dataset in which normal, wide, and double labels capture reproducible extremes of a broader morphological continuum. Across more than thirteen million accepted EODs, these recordings provide a first large-scale characterization of waveform variation and allow rare pulse forms to be studied across several years. The feeding analysis shows a clear increase in overall pulse rate around scheduled feeding, including a rise before the logged feeding time, while wide and double pulses follow a similar peri-feeding pattern at lower absolute rates. Coarse localisation reveals non-uniform use of the tank and a movement toward the dark boundary during late 2025.

The most interesting pattern is the increase in overall and special-pulse activity during late 2025. Because the strongest increase is concentrated in one period rather than recurring annually, a simple seasonal explanation is not sufficient. Its coincidence with a shift in coarse space use and unusual observations of increased proximity between the animals raises the possibility that the eels entered a different social state during this period. Given the limited knowledge of electric eel reproduction and the association of reproduction with structured, dark nest habitats (Assunção and Schwassmann 1995; Bastos 2020), a reproductive interpretation is a particularly interesting hypothesis. However, the present data cannot demonstrate mating, courtship, or another specific reproductive behaviour. The observations should therefore be regarded as evidence supporting a hypothesis that can be tested in future work, rather than as a demonstrated function of the special pulse classes.

Several questions remain unresolved. The waveform labels should not yet be interpreted as behavioural categories, and the double pulses in particular require recordings linked to individual animals, amplitude, and behaviour before they can be related to hunting or other specific actions. A related next experiment would be to relate special-pulse labels to high- versus low-voltage discharges explicitly: because the present classification ignores amplitude, it remains open whether wide or double shapes are enriched in one voltage regime, or whether their occurrence is mainly context-dependent rather than voltage-dependent. The cause of the long-term decline in overall pulse rate also remains unclear, as does the extent to which the narrow range of environmental variables in the Berlin tank constrains the observed patterns.

The most important next step is to apply the same classification and activity pipeline to the FLONA and Ducke recordings from the natural environment. The captive and field datasets should not be viewed as substitutes for one another. Rather, they provide complementary perspectives: captivity gave the recording duration and controlled context needed to discover these patterns; the field data are needed to determine their ecological meaning. Comparing the datasets will make it possible to distinguish patterns that are specific to the captive setting from those that are more likely to reflect the natural behaviour of *E. voltai*.

The main contribution of this thesis is therefore not that it resolves the social or reproductive behaviour of electric eels. Instead, it shows that long-term passive EOD recordings can reveal previously uncharacterized waveform variation and changes in activity that are difficult to detect through visual observation alone. The Berlin dataset provides the long-term reference needed to identify such changes, while comparison with recordings from the natural environment provides the next step toward understanding what these electrical patterns mean for the behaviour and ecology of *E. voltai*.

8 Appendix

8.1 List of Abbreviations

ddRAD	Double-digest restriction-site associated DNA
EOD	Electric organ discharge
FLONA	Floresta Nacional (national forest)
INPA	Instituto Nacional de Pesquisas da Amazônia
PCA	Principal component analysis
RF	Random forest
ROI	Region of interest

8.2 List of Figures

8.3 List of Tables

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References

Ardanaz, J., Silva, A., & Macadar, O. (2001). Temperature sensitivity of the electric organ discharge waveform in *Gymnotus carapo*. *Journal of Comparative Physiology A*, *187*(11), 853–864. https://doi.org/10.1007/s00359-001-0256-8

Assunção, M., & Schwassmann, H. (1995). Reproduction and larval development of electrophorus electricus on marajó island (pará, brazil). *Ichthyological Exploration of Freshwaters*, *6*(2), 175–184.

Bárbara, V. F., Cunha, A. C. d., Rodrigues, A. S. d. L., & Siqueira, E. Q. d. (2010). Monitoramento sazonal da qualidade da água do rio Araguari/AP. *Revista Biociências*, *16*(1), 57–72.

Bastos, D. A., Zuanon, J., Rapp Py-Daniel, L., & de Santana, C. D. (2021). Social predation in electric eels. *Ecology and evolution*, *11*(3), 1088–1092.

Bastos, D. A. (2020). *História natural de poraqués (Electrophorus spp.), gymnotiformes: Gymnotidae* [PhD thesis]. Instituto Nacional de Pesquisas da Amazônia (INPA) [Programa de Pós-Graduação em Biologia de Água Doce e Pesca Interior].

Bessa, E., Brandão, M. L., & Gonçalves-de-Freitas, E. (2022). Integrative approach on the diversity of nesting behaviour in fishes. *Fish and Fisheries*, *23*(3), 564–583.

Catania, K. (2014). The shocking predatory strike of the electric eel. *Science*, *346*(6214), 1231–1234.

Catania, K. C. (2015a). Electric eels concentrate their electric field to induce involuntary fatigue in struggling prey. *Current Biology*, *25*(22), 2889–2898.

Catania, K. C. (2015b). Electric eels use high-voltage to track fast-moving prey. *Nature Communications*, *6*(1), 8638.

Catania, K. C. (2017). Electrical potential of leaping eels. *Brain, Behavior and Evolution*, *89*(4), 262–273. https://doi.org/10.1159/000475743

Catania, K. C. (2019). The astonishing behavior of electric eels. *Frontiers in Integrative Neuroscience*, *13*, 23.

Crampton, W. G., & Hopkins, C. D. (2005). Nesting and paternal care in the weakly electric fish gymnotus (gymnotiformes: Gymnotidae) with descriptions of larval and adult electric organ discharges of two species. *Ichthyology & Herpetology*, *2005*(1), 48–60.

De Santana, C., Crampton, W., Dillman, C., Frederico, R., Sabaj, M., Covain, R., Ready, J., Zuanon, J., Oliveira, R., Mendes-Júnior, R., et al. (2019). Unexpected species diversity in electric eels with a description of the strongest living bioelectricity generator. *Nature Communications*, *10: 4000*.

de Santana, C. D. (2025). *From rapids to floodplains: Ecological boundaries drive genomic divergence without morphological change in electric eels* [Preliminary report of ddRAD analyses (preprint; not yet published; year provisional)].

Ebrahim, M. A., Ragab, A. S., Aziz, B. A., & AbdelHadi, H. A. (2025). Electric eel foraging optimization based control design of islanded microgrid. *Scientific Reports*, *15*, 8144. https://doi.org/10.1038/s41598-025-91006-y

Hostache, G., & Mol, J. H. (1998). Reproductive biology of the neotropical armoured catfish hoplosternum littorale (siluriformes—callichthyidae): A synthesis stressing the role of the floating bubble nest. *Aquatic Living Resources*, *11*(3), 173–185.

Ilha, P., Schiesari, L., Yanagawa, F. I., Jankowski, K., & Navas, C. A. (2018). Deforestation and stream warming affect body size of Amazonian fishes. *PLOS ONE*, *13*(5), e0196560. https://doi.org/10.1371/journal.pone.0196560

Johannsson, O. E., Nascimento, T. L., Agasild, H., Zingel, P., Duarte, R. M., de Boeck, G., Cremazy, A., da Silva, J. M., Morris, C., Wood, C. M., & Val, A. L. (2026). Amazonian drought of 2023: Environmental conditions relevant to fishes. *Journal of Fish Biology*, *108*(3), 803–823. https://doi.org/10.1111/jfb.70313

Johansen, K., Lenfant, C., Schmidt-Nielsen, K., & Petersen, J. A. (1968). Gas exchange and control of breathing in the electric eel, electrophorus electricus. *Zeitschrift für vergleichende Physiologie*, *61*(2), 137–163.

Keynes, R., & Martins-Ferreira, H. (1953). Membrane potentials in the electroplates of the electric eel. *The Journal of physiology*, *119*(2-3), 315.

McAnelly, M. L., & Zakon, H. H. (2000). Coregulation of voltage-dependent kinetics of Na^+ and K^+ currents in electric organ. *Journal of Neuroscience*, *20*(9), 3408–3414.

Mendes-Junior, R. N. G., Sá-Oliveira, J. C., & Ferrari, S. F. (2016). Biology of the electric eel, electrophorus electricus, linnaeus, 1766 (gymnotiformes: Gymnotidae) on the floodplain of the curiaú river, eastern amazonia. *Reviews in Fish Biology and Fisheries*, *26*(1), 83–91.

Pei, H., Hu, H., Dong, Y., Zhu, H., Zhang, C., Zhou, Y., Huang, J., Shi, S., Wang, Z., Wu, X., et al. (2025). Electric eels inspired iontronic artificial skin with multimodal perception and in-sensor reservoir computing. *Advanced Functional Materials*, 2506431.

Röpke, C., Pires, T. H. S., Zuchi, N., Zuanon, J., & Amadio, S. (2022). Effects of climate-driven hydrological changes in the reproduction of Amazonian floodplain fishes. *Journal of Applied Ecology*, *59*(4), 1134–1145. https://doi.org/10.1111/1365-2664.14126

Sakaki, S., Ito, R., Abe, H., Kinoshita, M., Hondo, E., & Iida, A. (2023). Electric organ discharge from electric eel facilitates dna transformation into teleost larvae in laboratory conditions. *PeerJ*, *11*, e16596.

Sá-Oliveira, J. C., & Mendes Júnior, R. N. G. (2012). Fecundidade e tipo de desova do poraquê, *Electrophorus electricus*, (Linnaeus, 1766) (Osteichthyies: Gymnotiformes: Gymnotidae) da área de Proteção Ambiental - APA - do Rio Curiaú, Macapá-AP. *Biota Amazônia*, *2*(1), 32–36. https://doi.org/10.18561/2179-5746/biotaamazonia.v2n1p32-36

Senarat, S., Matsumoto, A., Nagasawa, T., Sakaki, S., Tsuzuki, D., Uchida, K., Kuwahara, M., Nikaido, M., Hondo, E., & Iida, A. (2025). Ventral-to-dorsal electrocyte development in electric organs of electric eel (electrophorus). *bioRxiv*. https://doi.org/10.1101/2024.08.21.606117

Silva, A., Quintana, L., Galeano, M., Errandonea, P., & Macadar, O. (1999). Water temperature sensitivity of EOD waveform in *Brachyhypopomus pinnicaudatus*. *Journal of Comparative Physiology A*, *185*(2), 187–197. https://doi.org/10.1007/s003590050377

Sun, H., Fu, X., Xie, S., Jiang, Y., & Peng, H. (2016). Electrochemical capacitors with high output voltages that mimic electric eels. *Advanced Materials*, *28*(10), 2070–2076.

Viana, L. F., Kummrow, F., Cardoso, C. A. L., de Lima, N. A., Solórzano, J. C. J., Crispim, B. d. A., Barufatti, A., & Florentino, A. C. (2021). High concentrations of metals in the waters from Araguari River lower section (Amazon biome): Relationship with land use and cover, ecotoxicological effects and risks to aquatic biota. *Chemosphere*, *285*, 131451. https://doi.org/10.1016/j.chemosphere.2021.131451

Volta, A. (1800). On the electricity excited by the mere contact of conducting substances of different kinds. *Philosophical Transactions of the Royal Society of London*, *90*, 403–431. https://doi.org/10.1098/rstl.1800.0018

Xu, J., Cui, X., & Zhang, H. (2021). The third form electric organ discharge of electric eels. *Scientific reports*, *11*(1), 6193.

Zhao, W., Wang, L., Zhang, Z., Fan, H., Zhang, J., Mirjalili, S., Khodadadi, N., & Cao, Q. (2024). Electric eel foraging optimization: A new bio-inspired optimizer for engineering applications. *Expert Systems with Applications*, *238*, 122200. https://doi.org/10.1016/j.eswa.2023.122200